

(12) **United States Patent**
Wang et al.

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(45) **Date of Patent:** ***Aug. 12, 2025**

(54) **SEMICONDUCTOR TRANSISTOR STRUCTURE WITH NANOSTRUCTURES AND CONDUCTIVE STRUCTURE AND METHOD FOR MANUFACTURING THE SAME**

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

This patent is subject to a terminal disclaimer.

(21) Appl. No.: **18/604,670**

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(65) **Prior Publication Data**

US 2024/0222433 A1 Jul. 4, 2024

Related U.S. Application Data

(63) Continuation of application No. 17/978,409, filed on Nov. 1, 2022, now Pat. No. 11,961,886, which is a
(Continued)

(51) **Int. Cl.**
H10D 62/10 (2025.01)
H10D 84/01 (2025.01)
H10D 84/03 (2025.01)

(52) **U.S. Cl.**
CPC **H10D 62/121** (2025.01); **H10D 84/013**
(2025.01); **H10D 84/0149** (2025.01); **H10D**
84/038 (2025.01)

(58) **Field of Classification Search**
CPC H01L 29/0673; H01L 21/823418; H01L
21/823475; H01L 23/5226; H10D 62/121;
(Continued)

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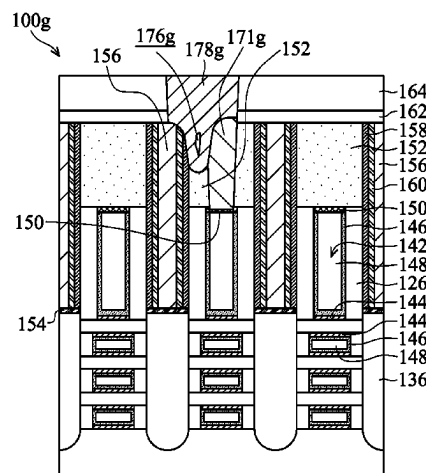
Primary Examiner — Changhyun Yi

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(57) **ABSTRACT**

Semiconductor structures and methods for manufacturing the same are provided. The semiconductor structure includes nanostructures spaced apart from each other in a first direction and a gate structure formed over and around the nanostructures. The semiconductor structure further includes a gate spacer covering a sidewall of the gate structure and a source/drain structure attached to the nanostructures in a second direction. The semiconductor structure further includes a contact spaced apart from the gate structure by the gate spacer in the second direction and a first

(Continued)



conductive structure landing over the gate structure. The semiconductor structure further includes a second conductive structure formed over the gate spacer. In addition, a portion of the second conductive structure is sandwiched between the first conductive structure and the contact.

20 Claims, 29 Drawing Sheets

Related U.S. Application Data

continuation of application No. 17/325,419, filed on May 20, 2021, now Pat. No. 11,527,614.

(60) Provisional application No. 63/158,489, filed on Mar. 9, 2021.

(58) Field of Classification Search

CPC H10D 84/013; H10D 84/0149;
H10D 84/038

See application file for complete search history.

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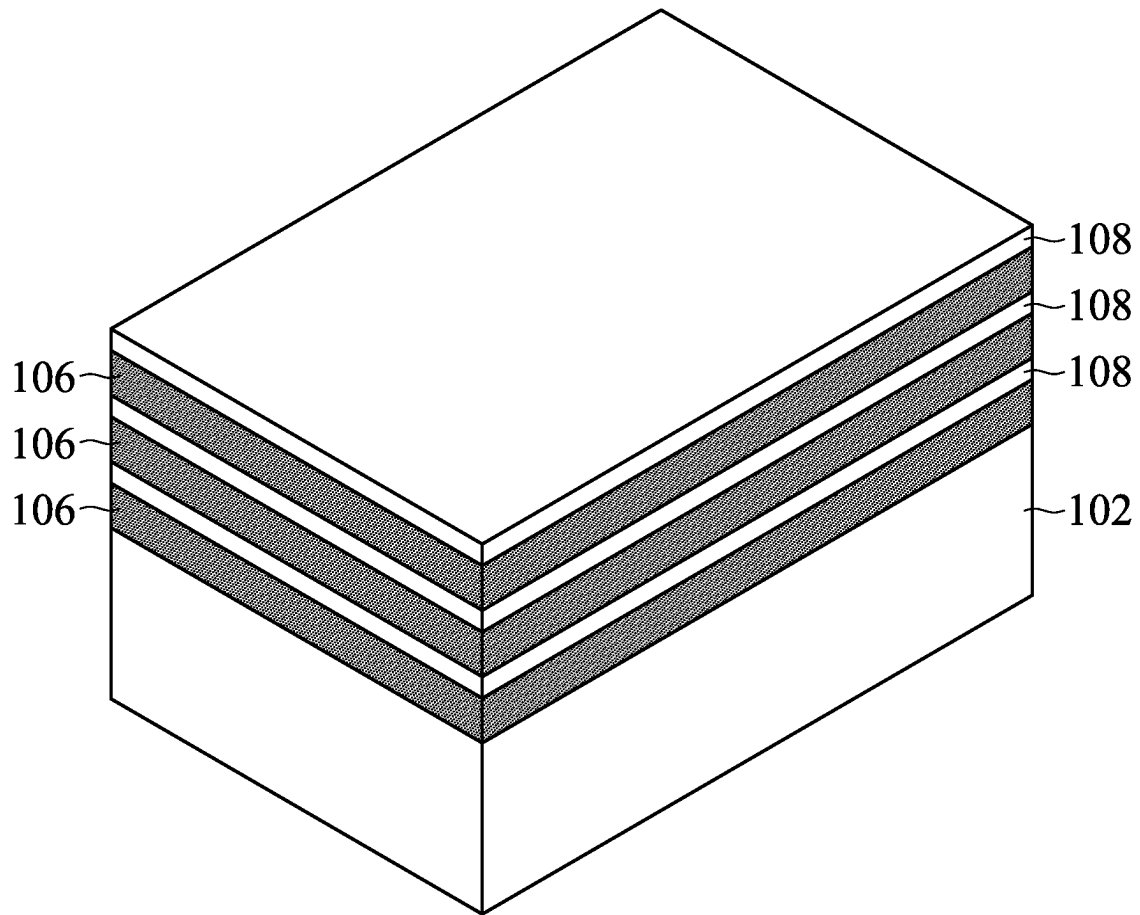


FIG. 1A

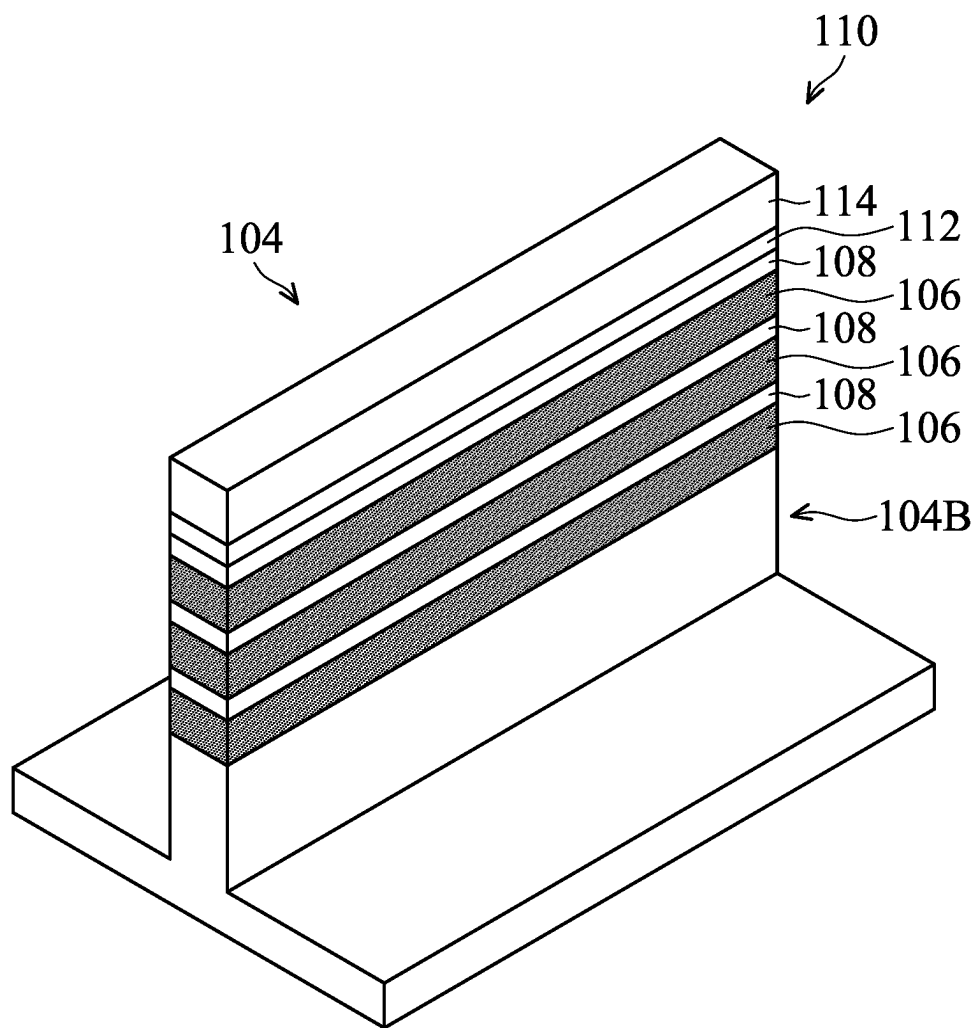


FIG. 1B

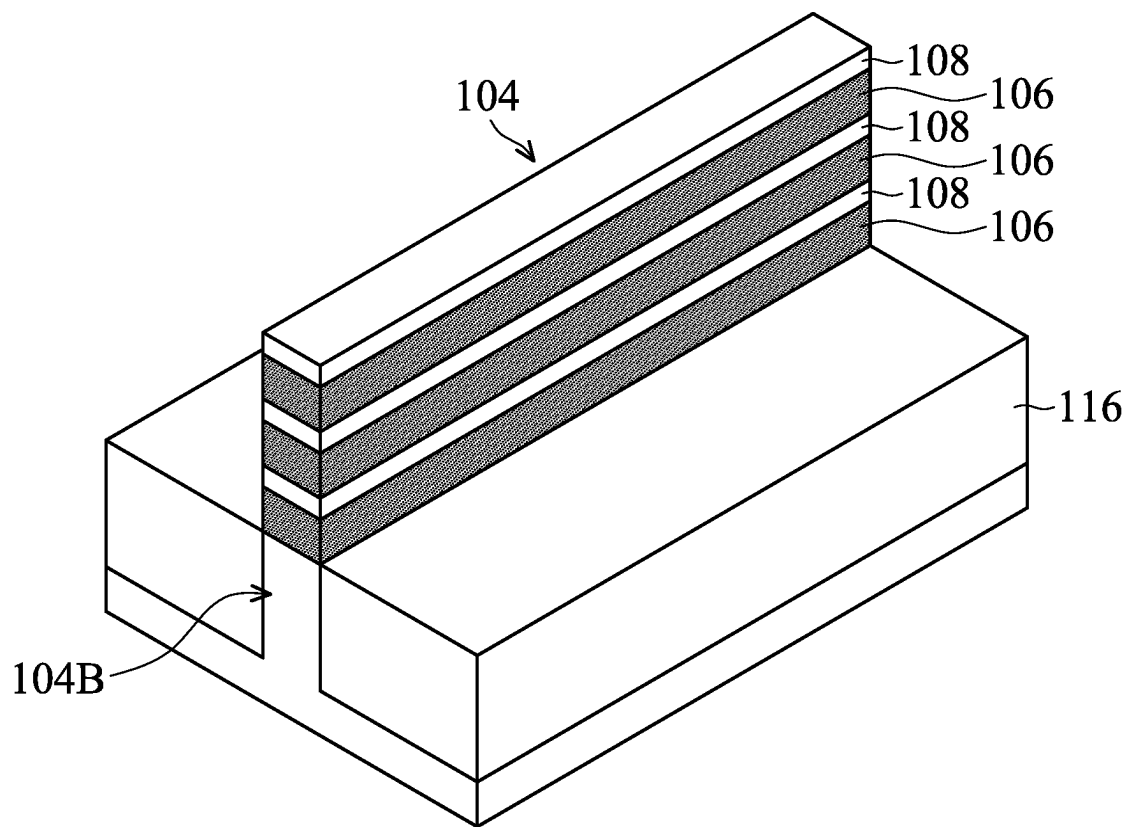


FIG. 1C

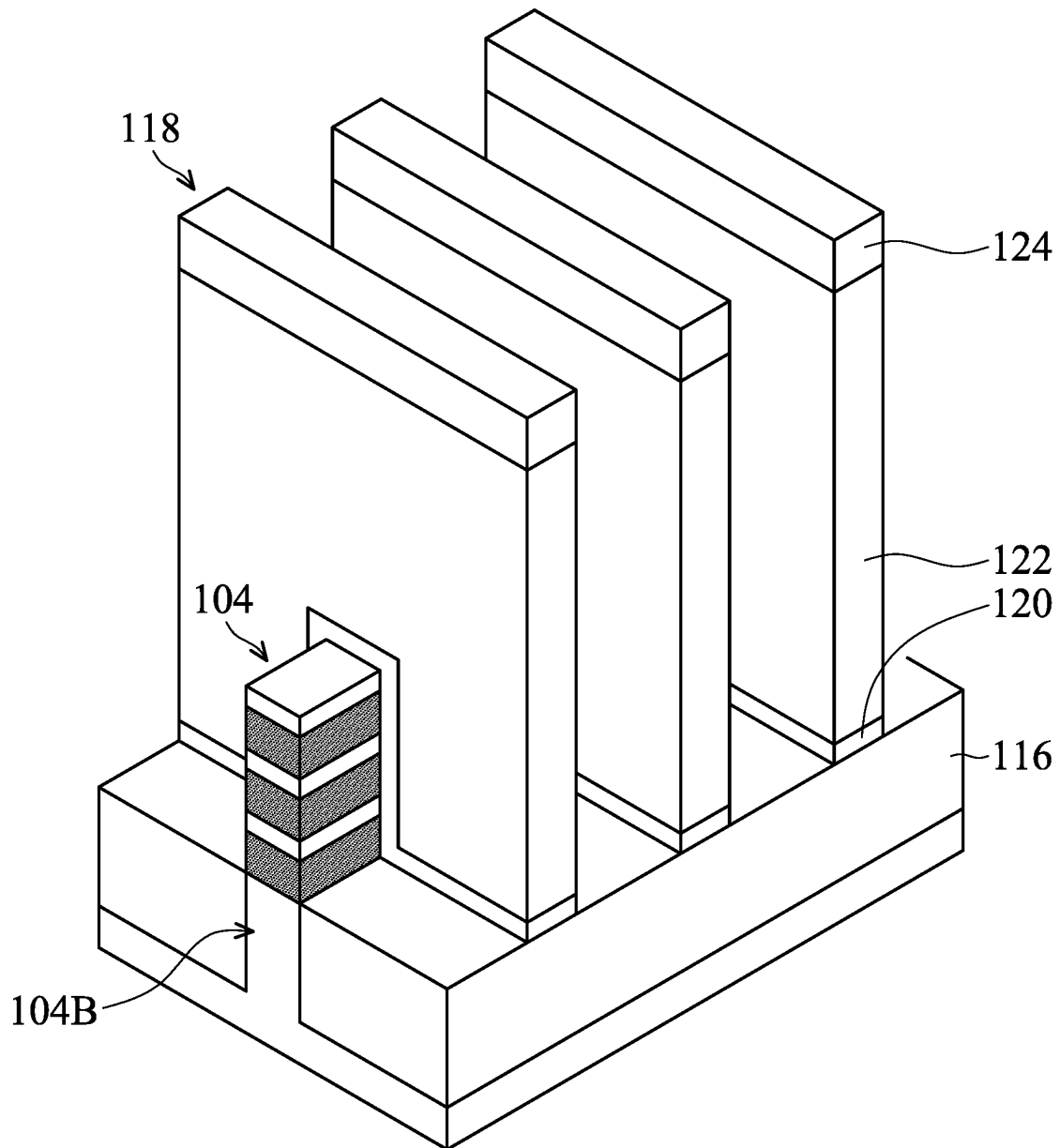


FIG. 1D

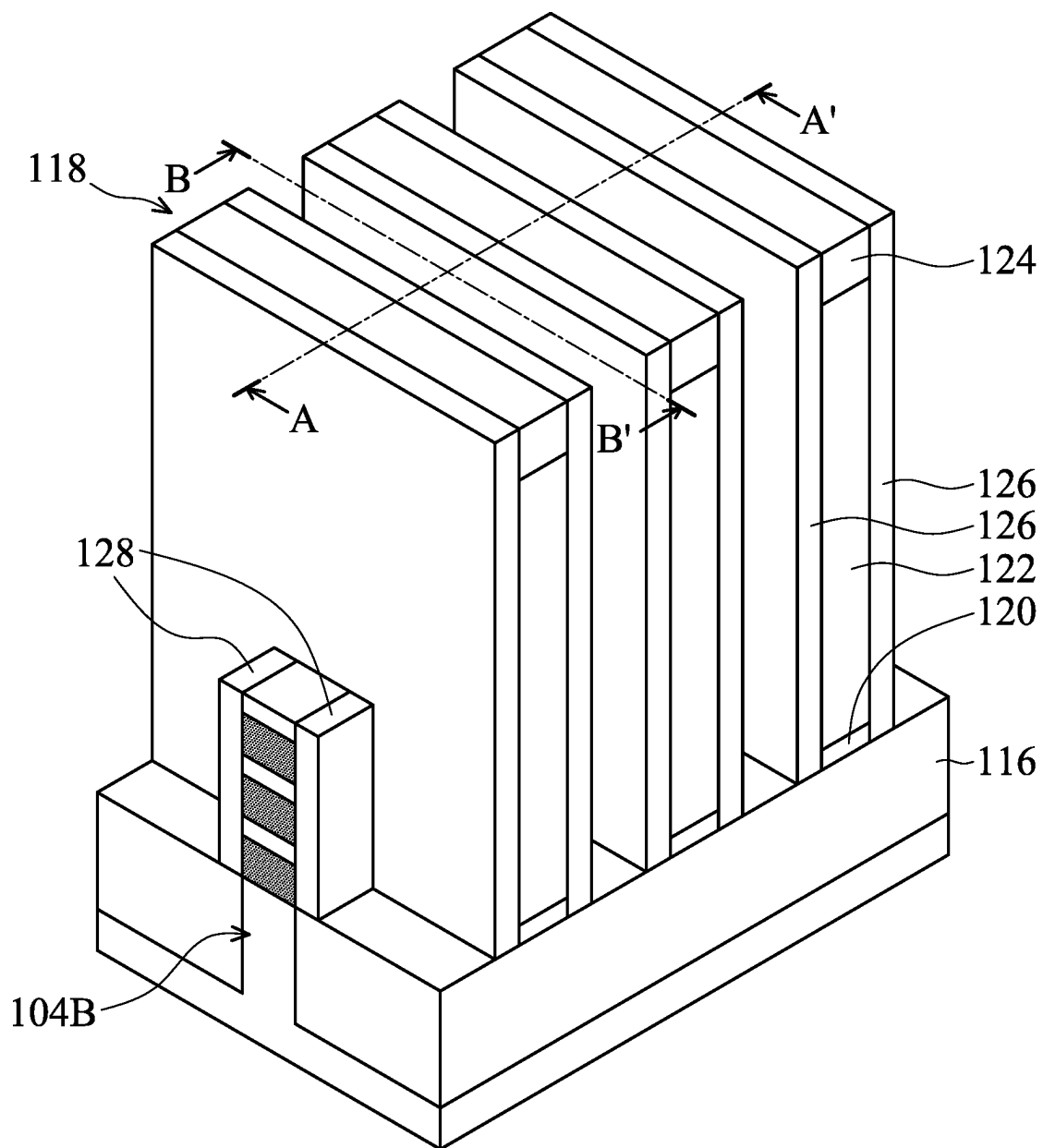


FIG. 1E

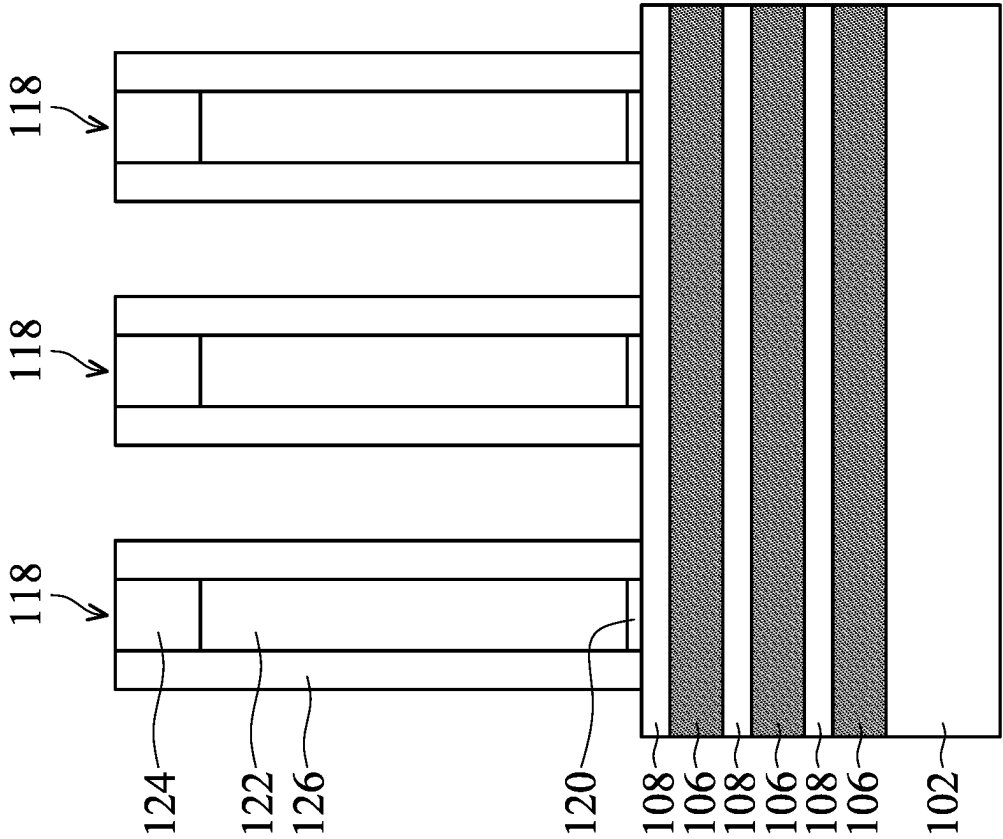


FIG. 2A-1

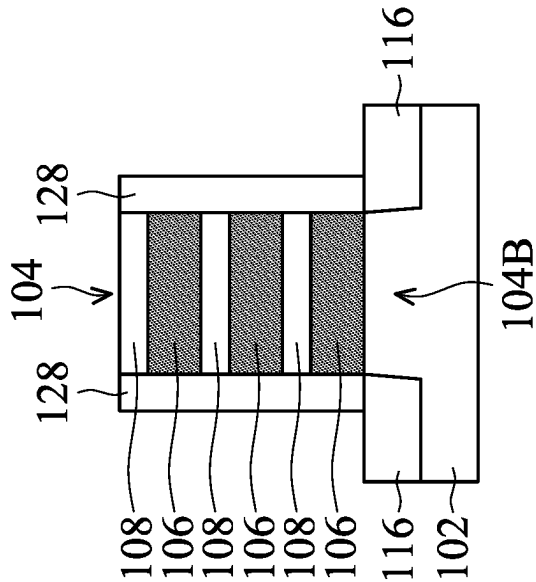


FIG. 2A-2

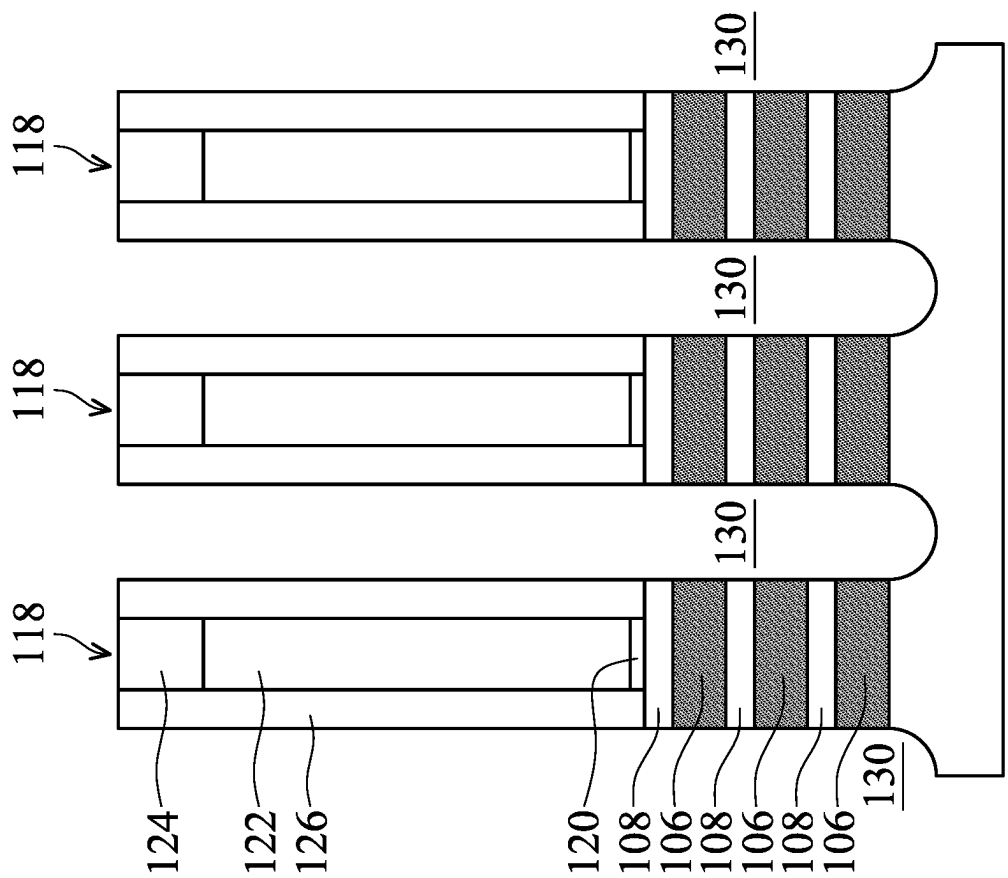


FIG. 2B-1

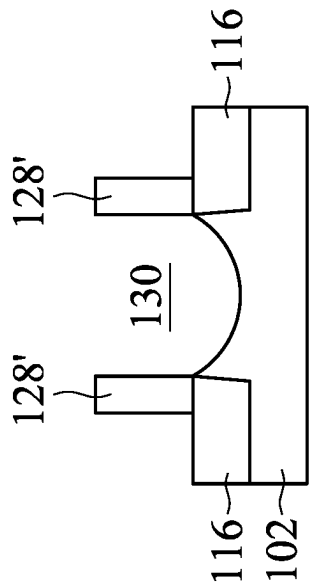


FIG. 2B-2

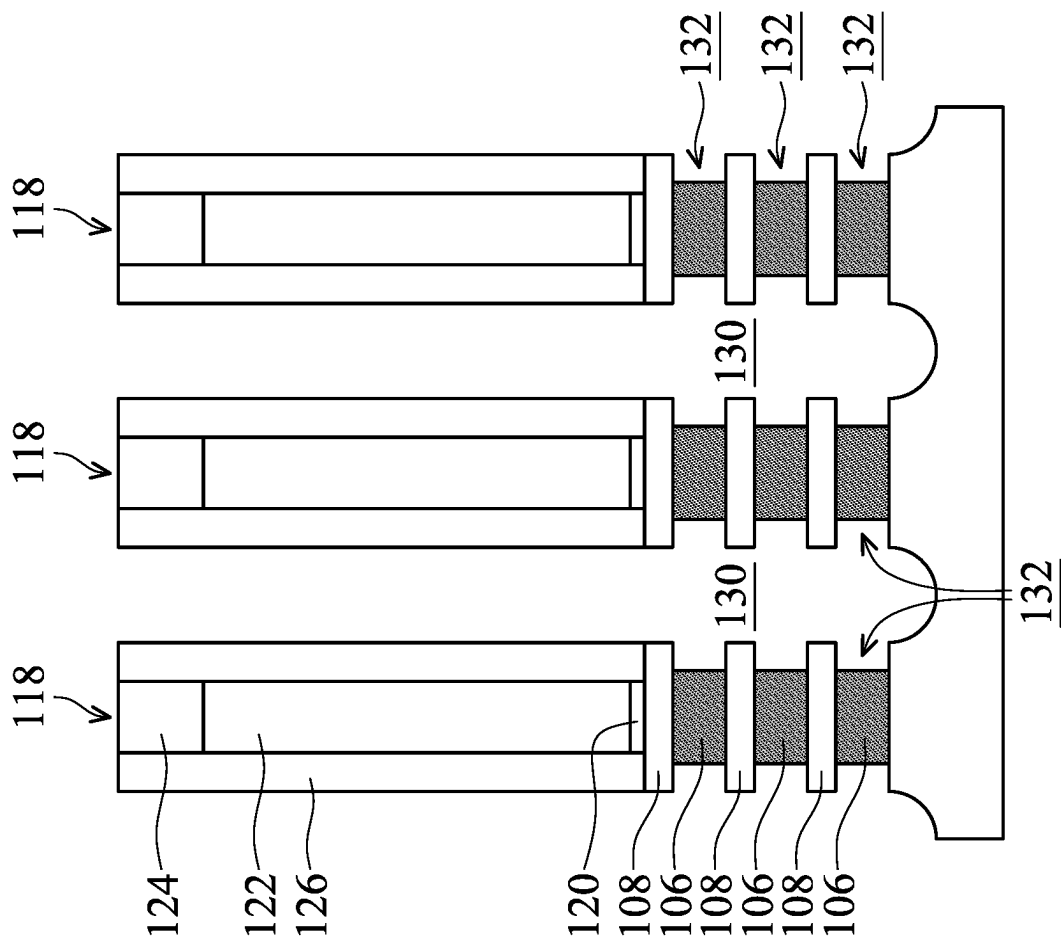


FIG. 2C-1

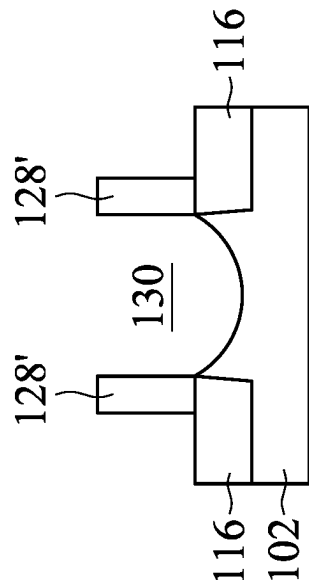


FIG. 2C-2



FIG. 2D-1

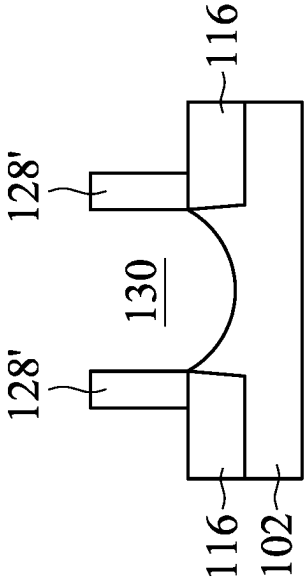


FIG. 2D-2

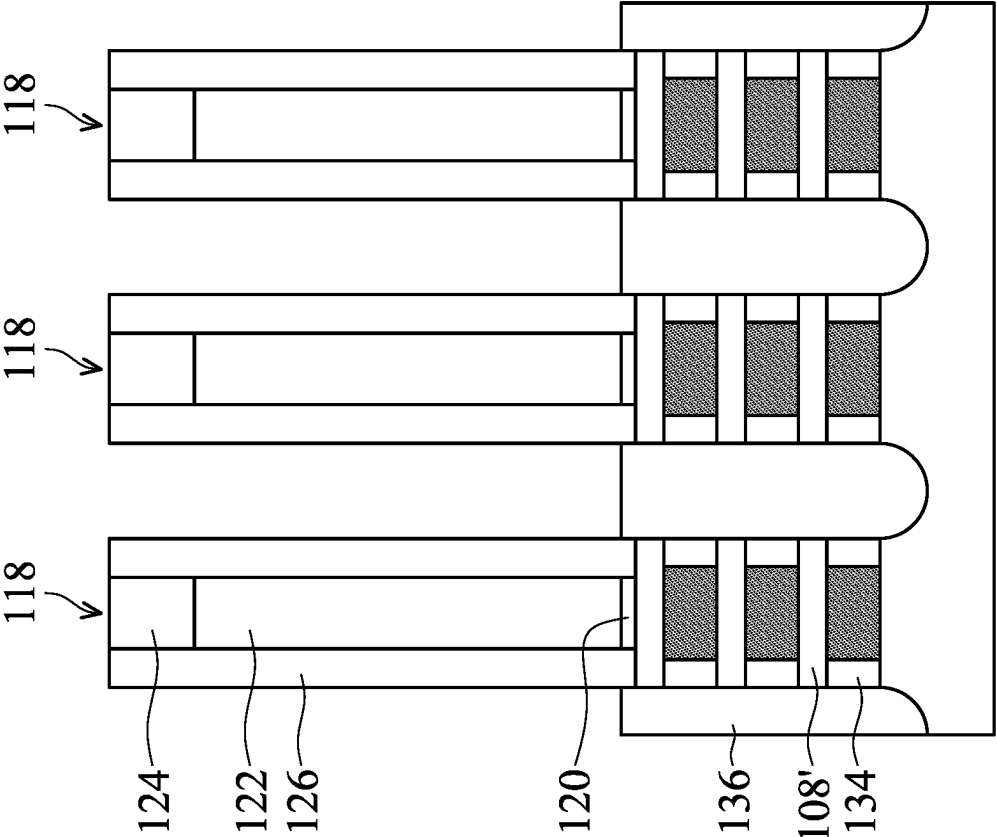


FIG. 2E-1

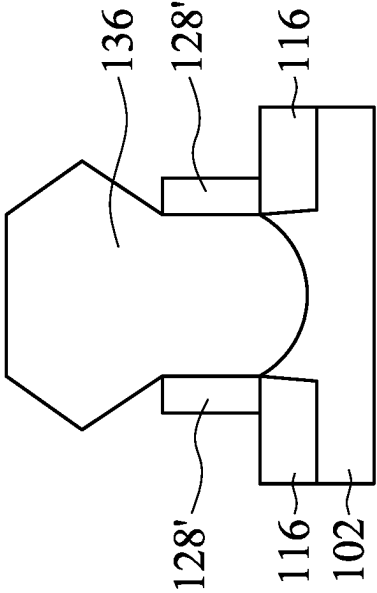


FIG. 2E-2

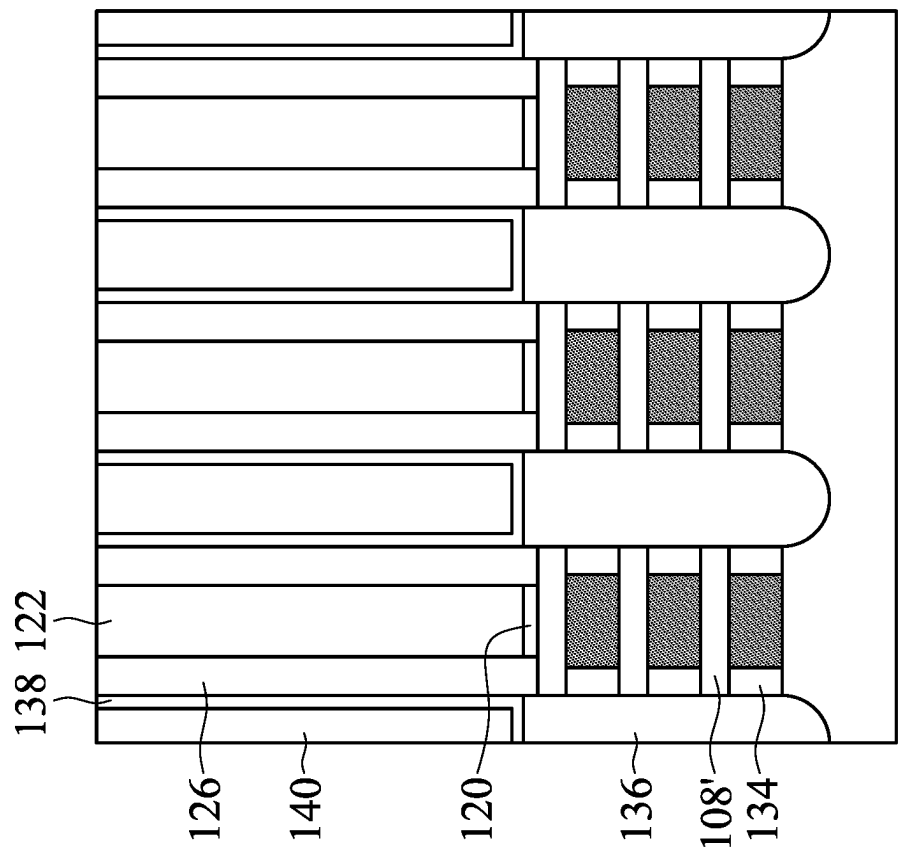


FIG. 2F-1

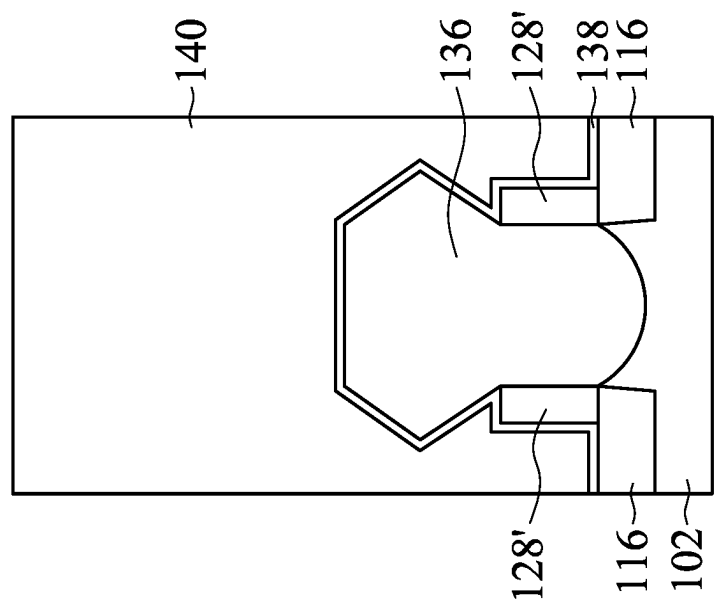


FIG. 2F-2

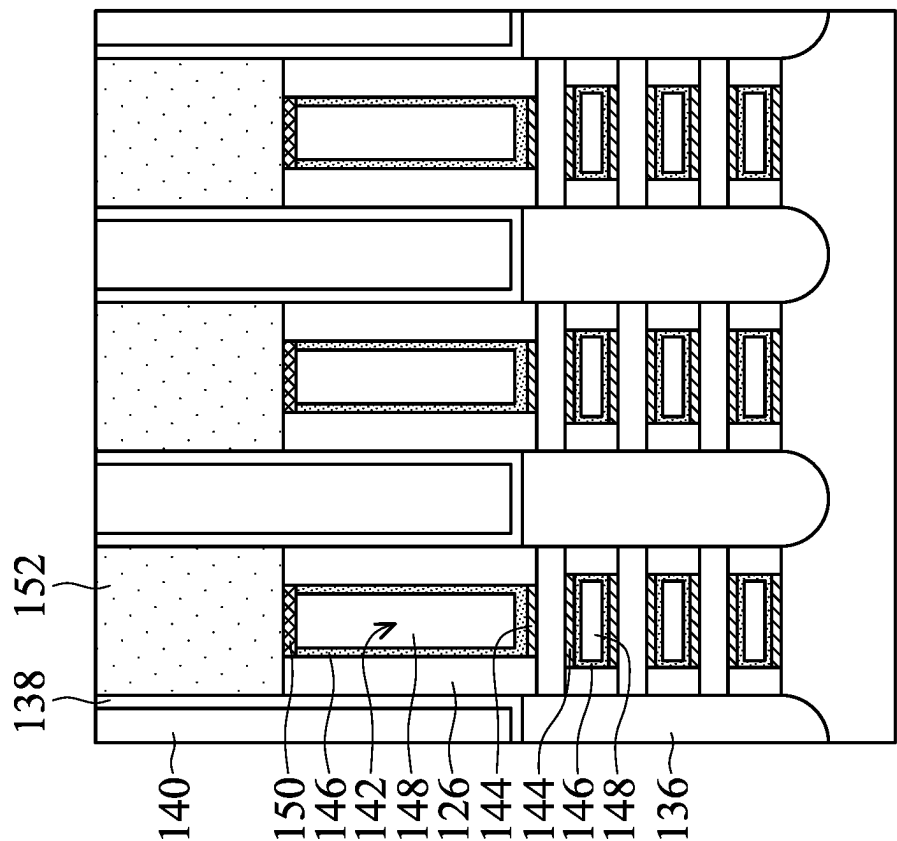


FIG. 2G-1

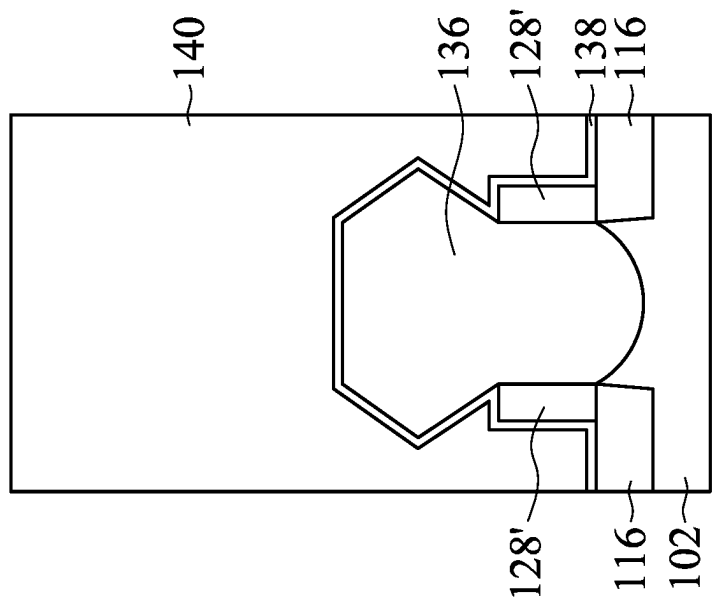


FIG. 2G-2

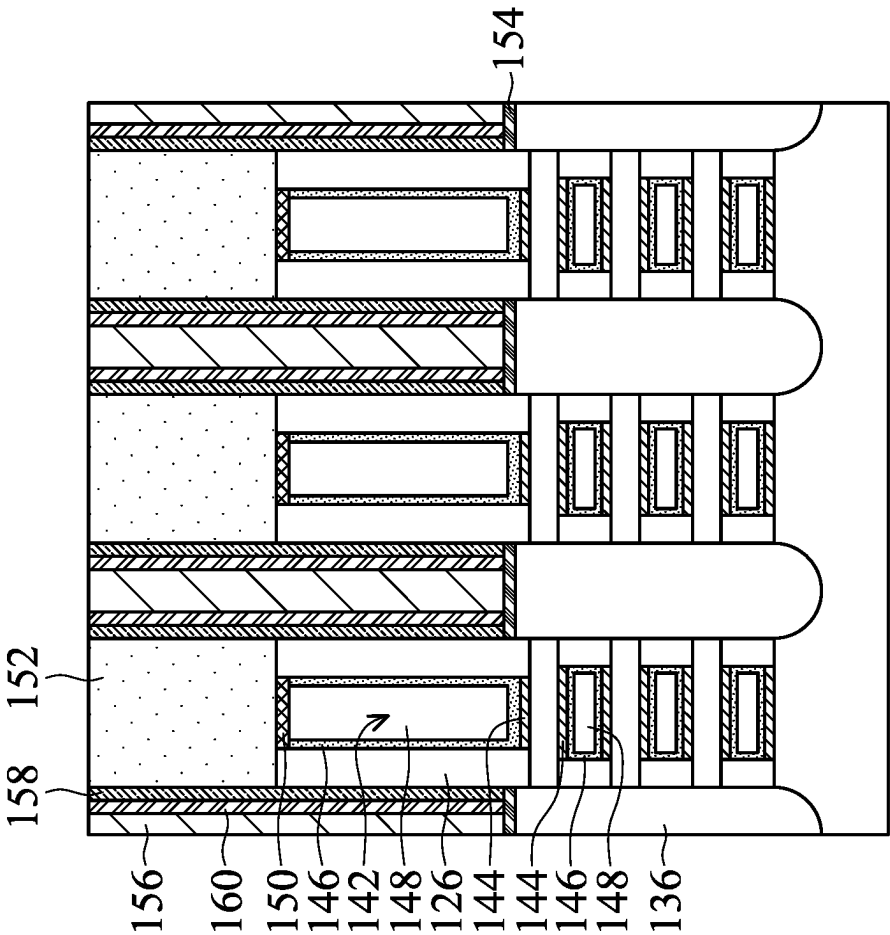


FIG. 2H-1

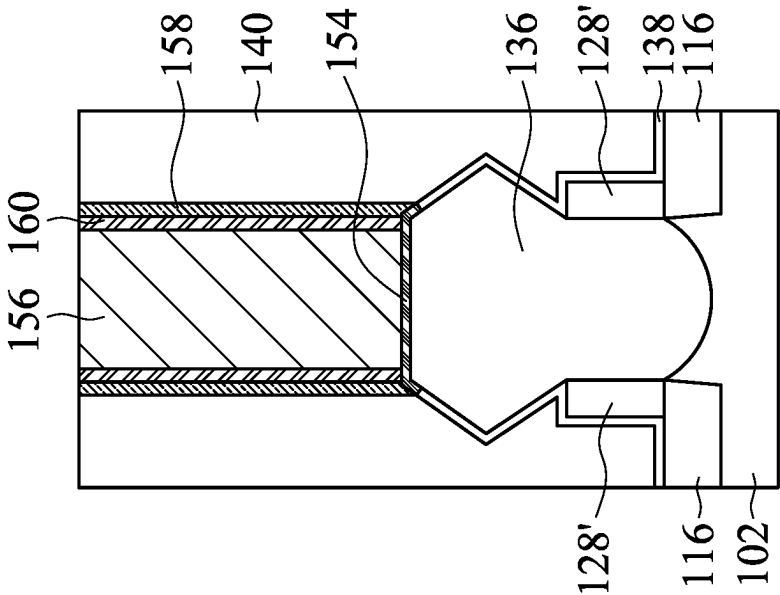


FIG. 2H-2

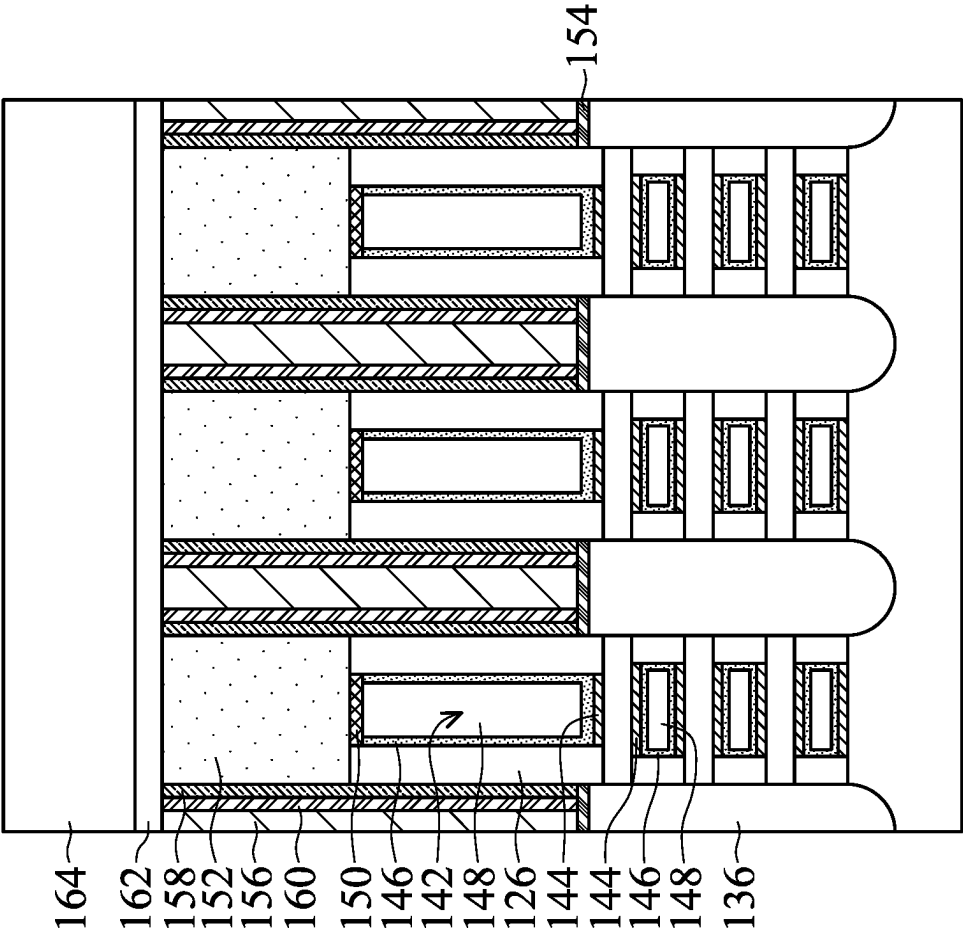


FIG. 2I-1

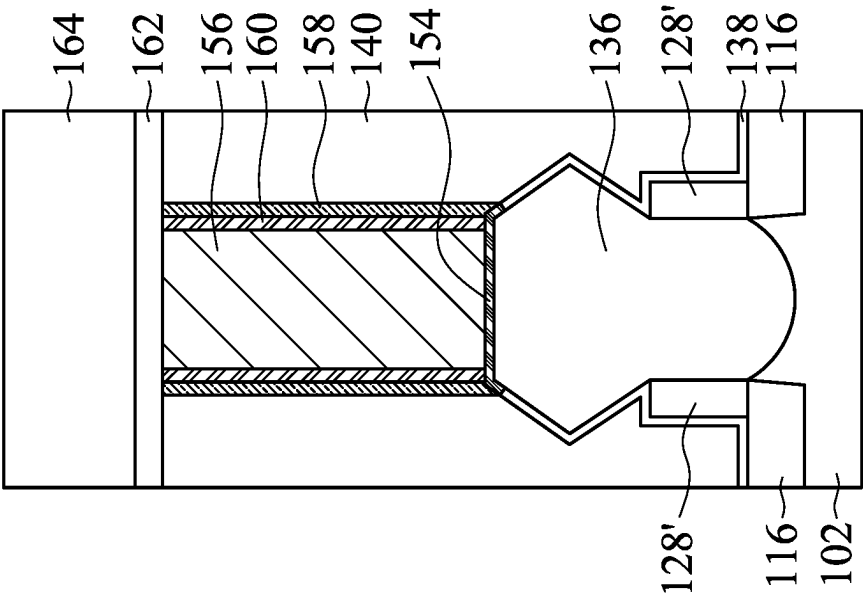


FIG. 2I-2

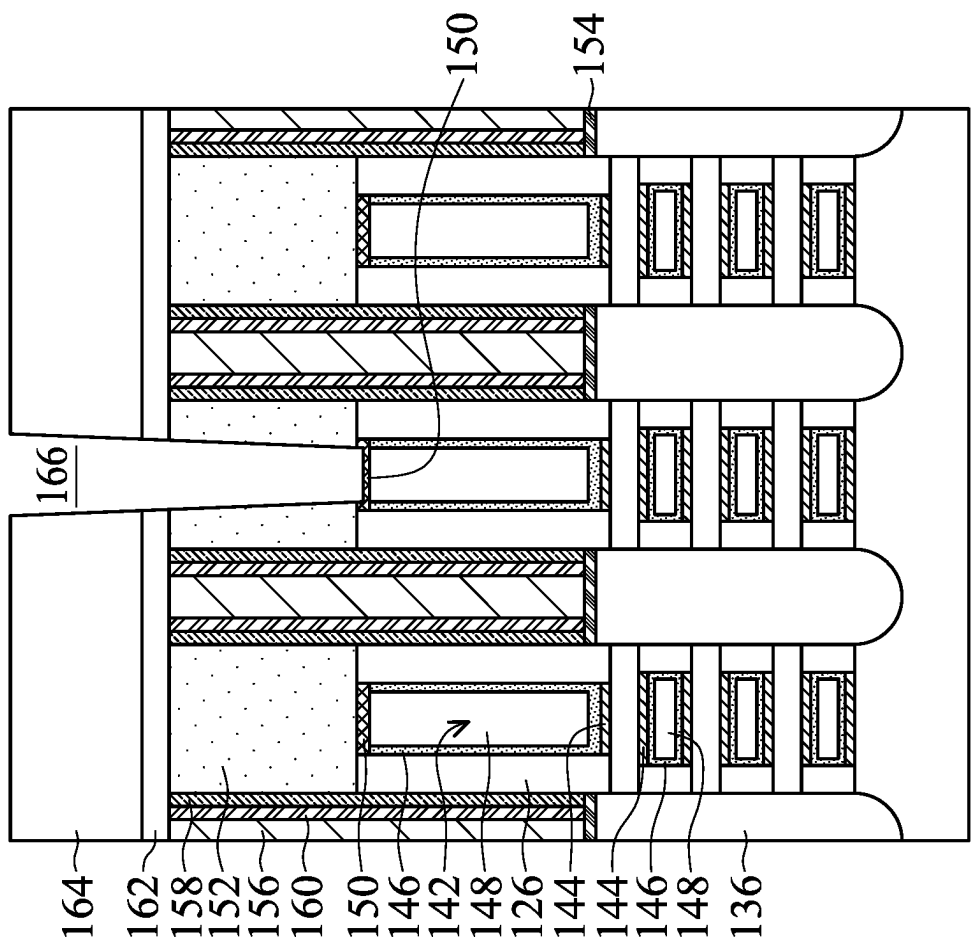


FIG. 2J-1

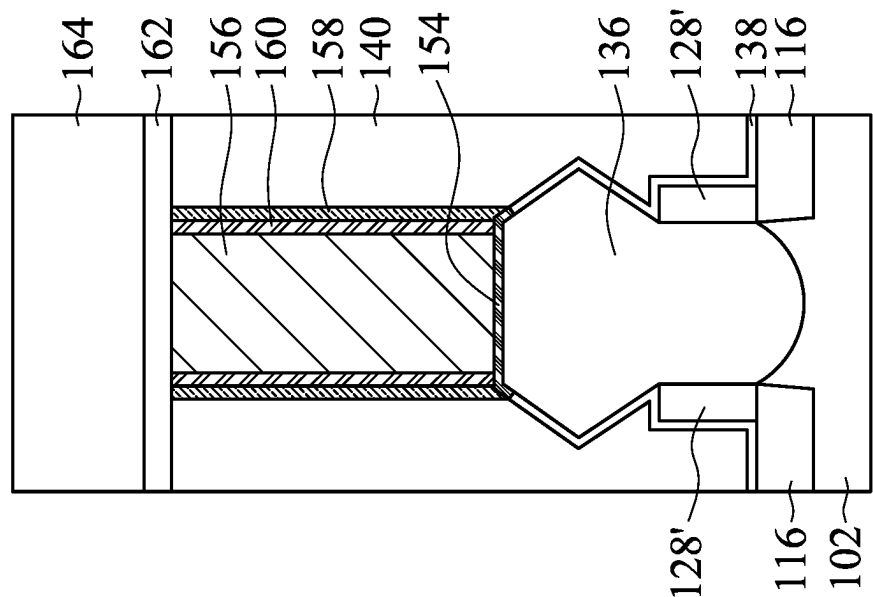


FIG. 2J-2

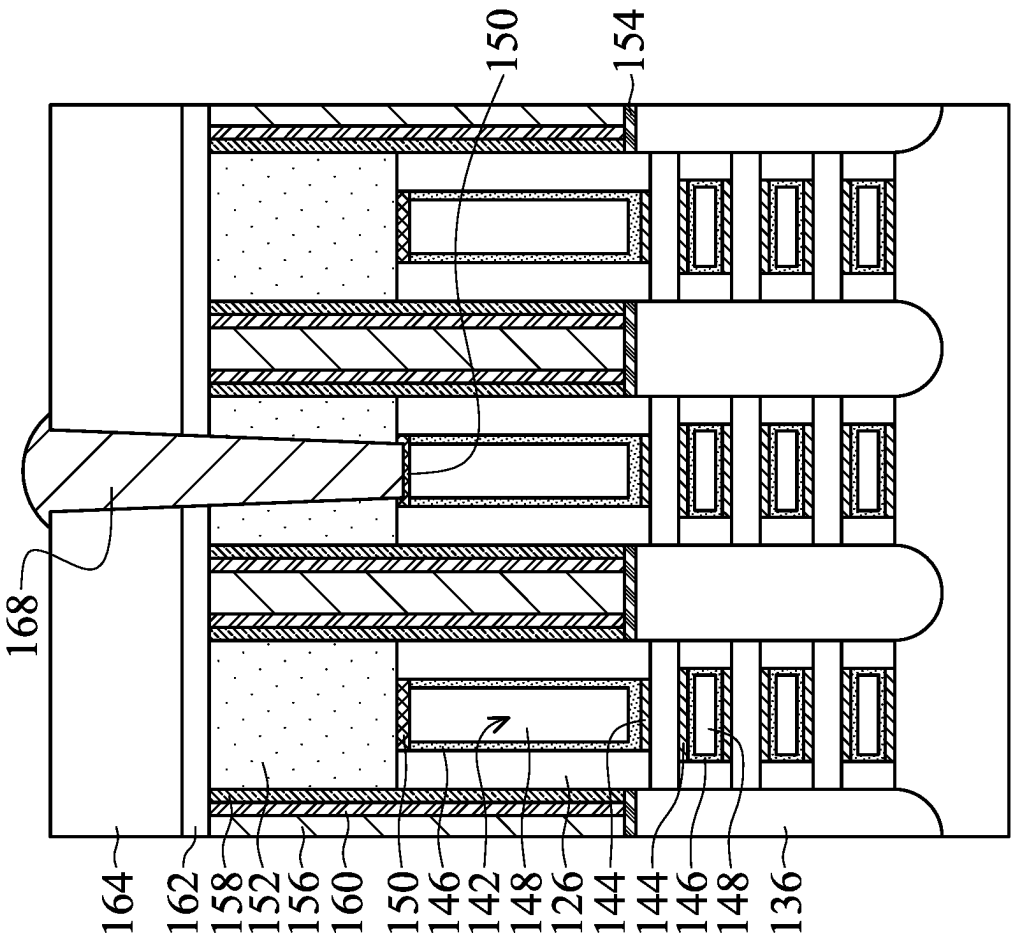


FIG. 2K-1

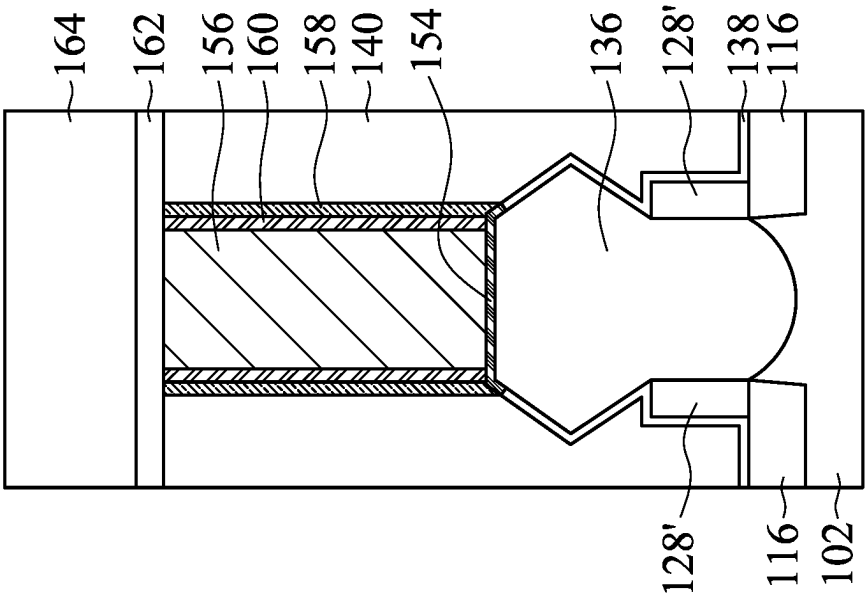


FIG. 2K-2

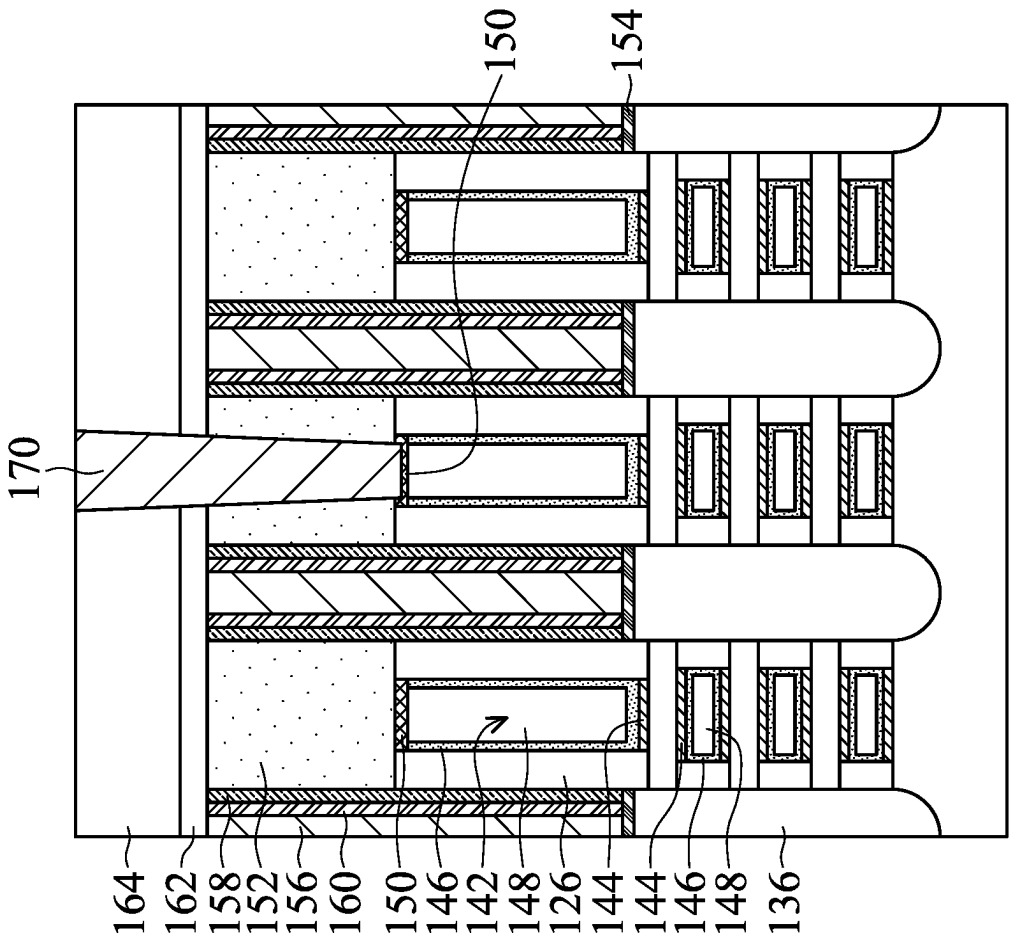


FIG. 2L-1

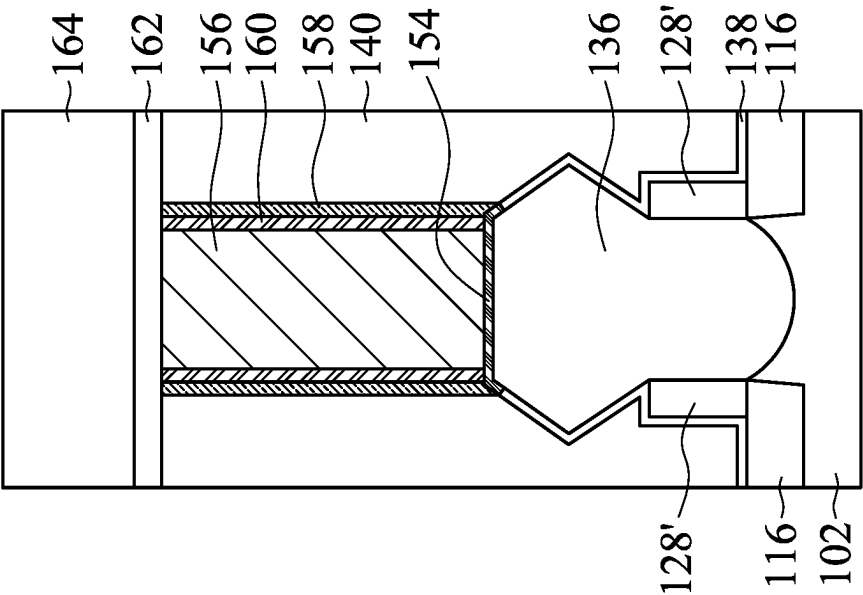


FIG. 2L-2

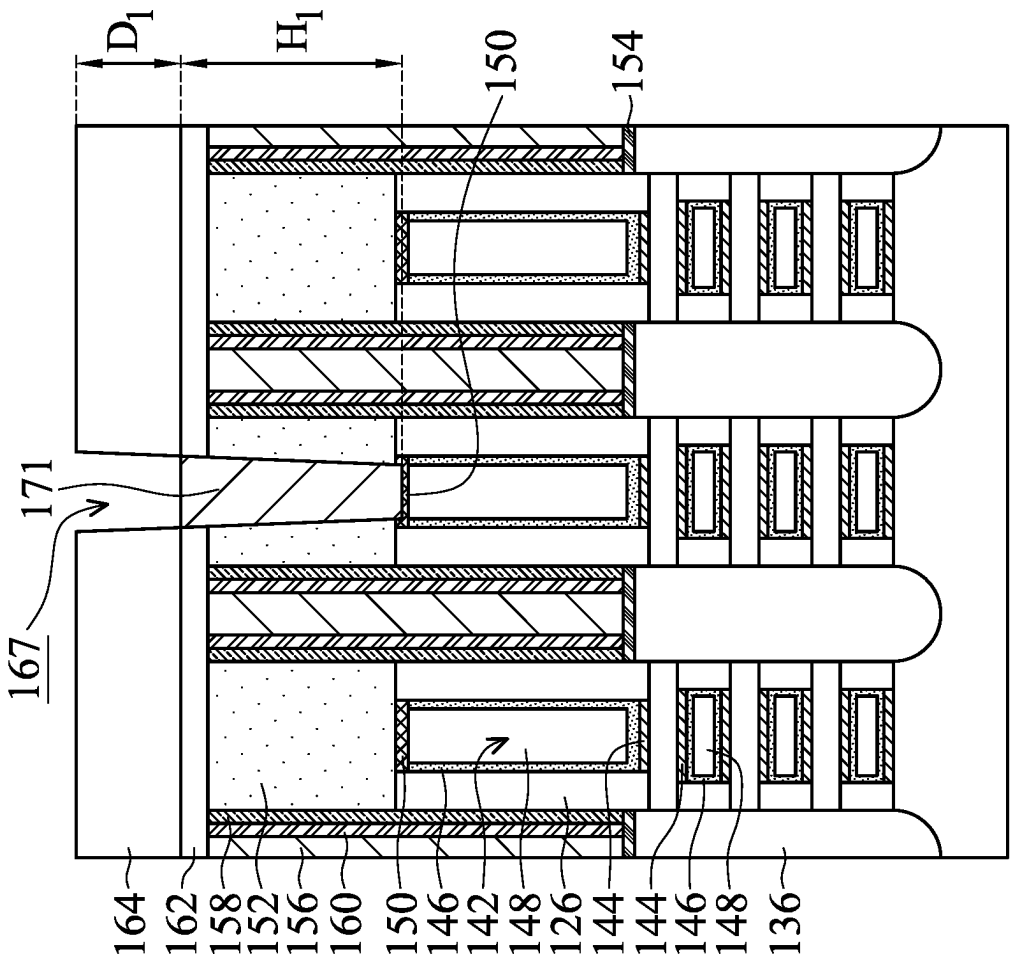


FIG. 2M-1

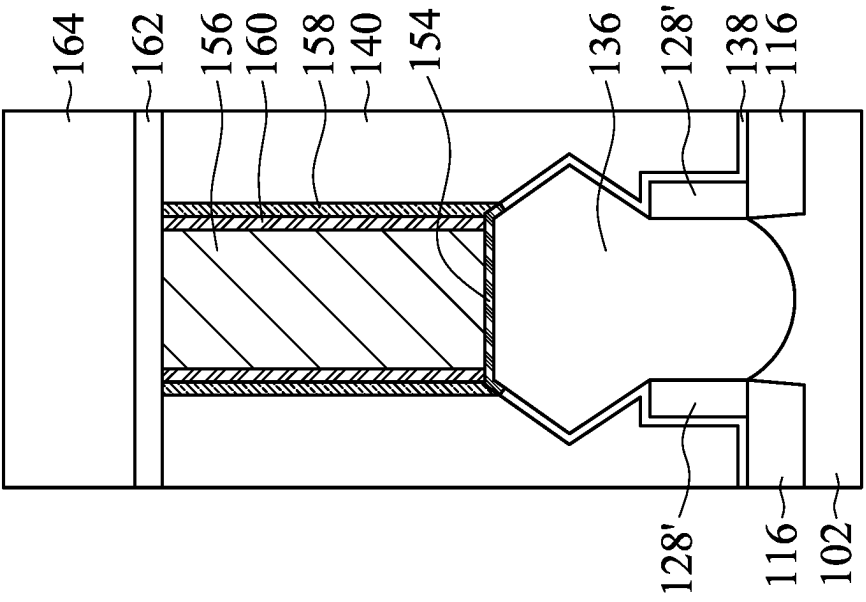


FIG. 2M-2

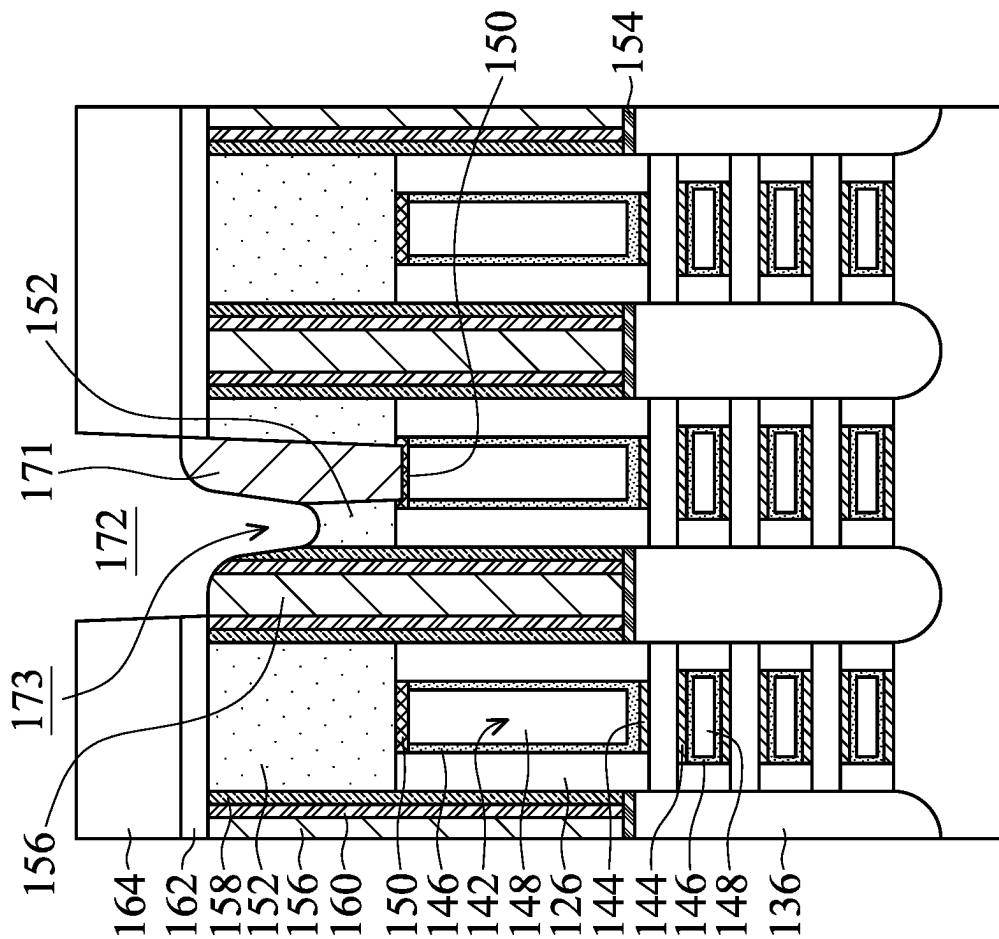


FIG. 2N-1

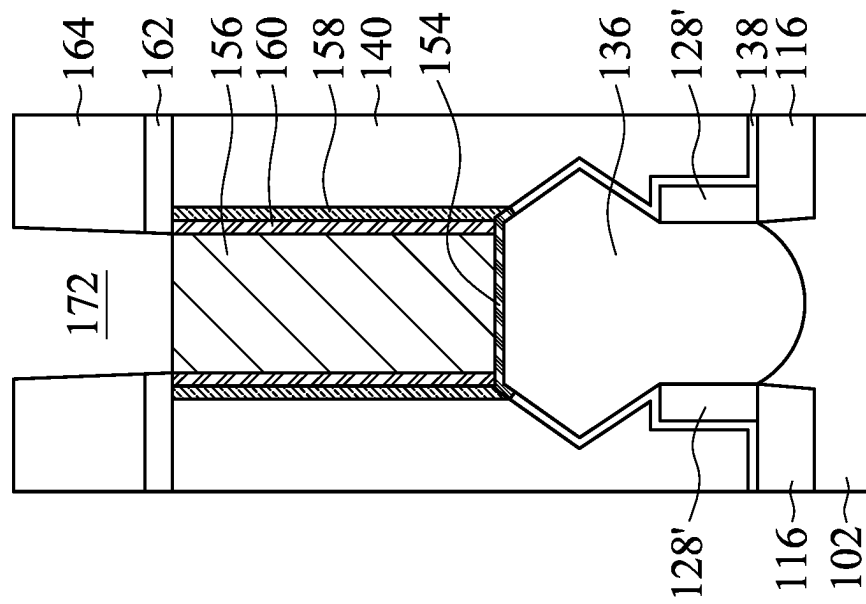


FIG. 2N-2

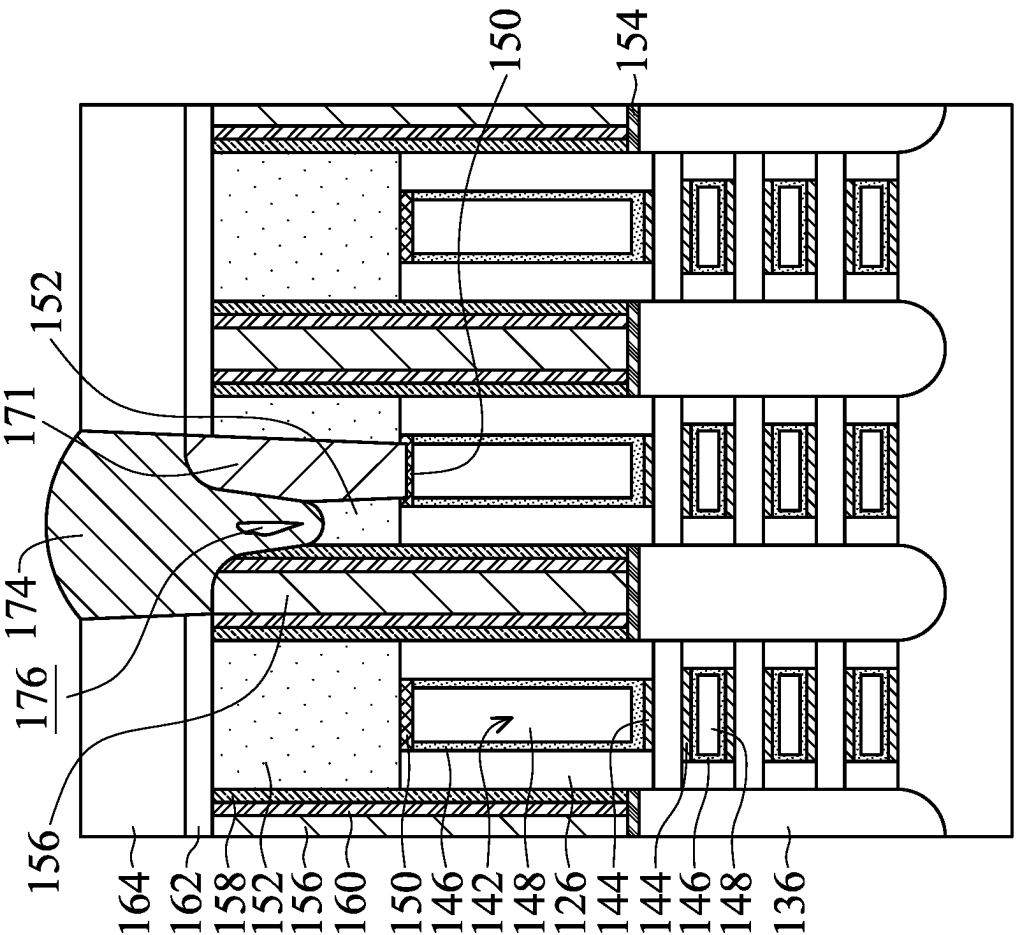


FIG. 20-1

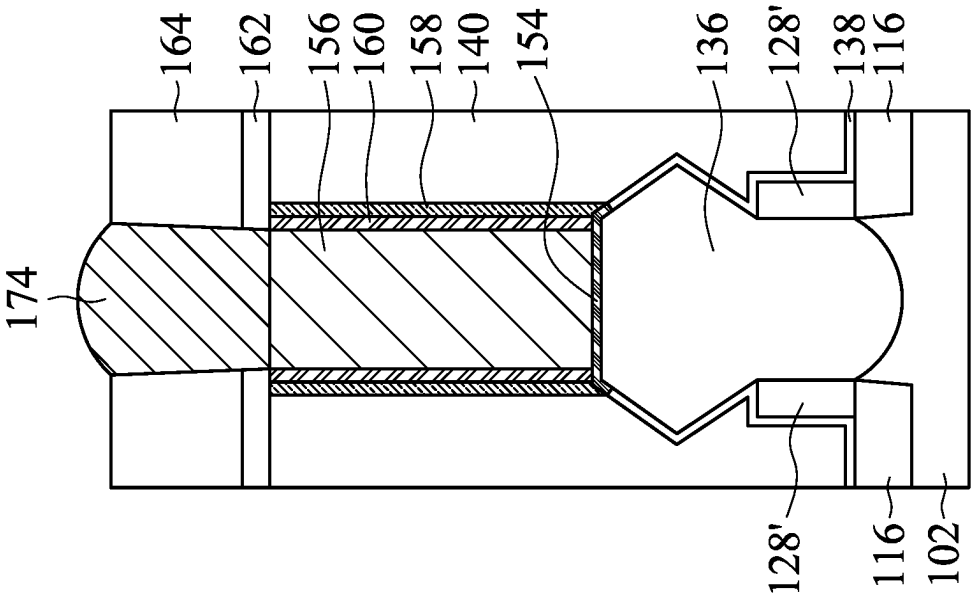


FIG. 20-2

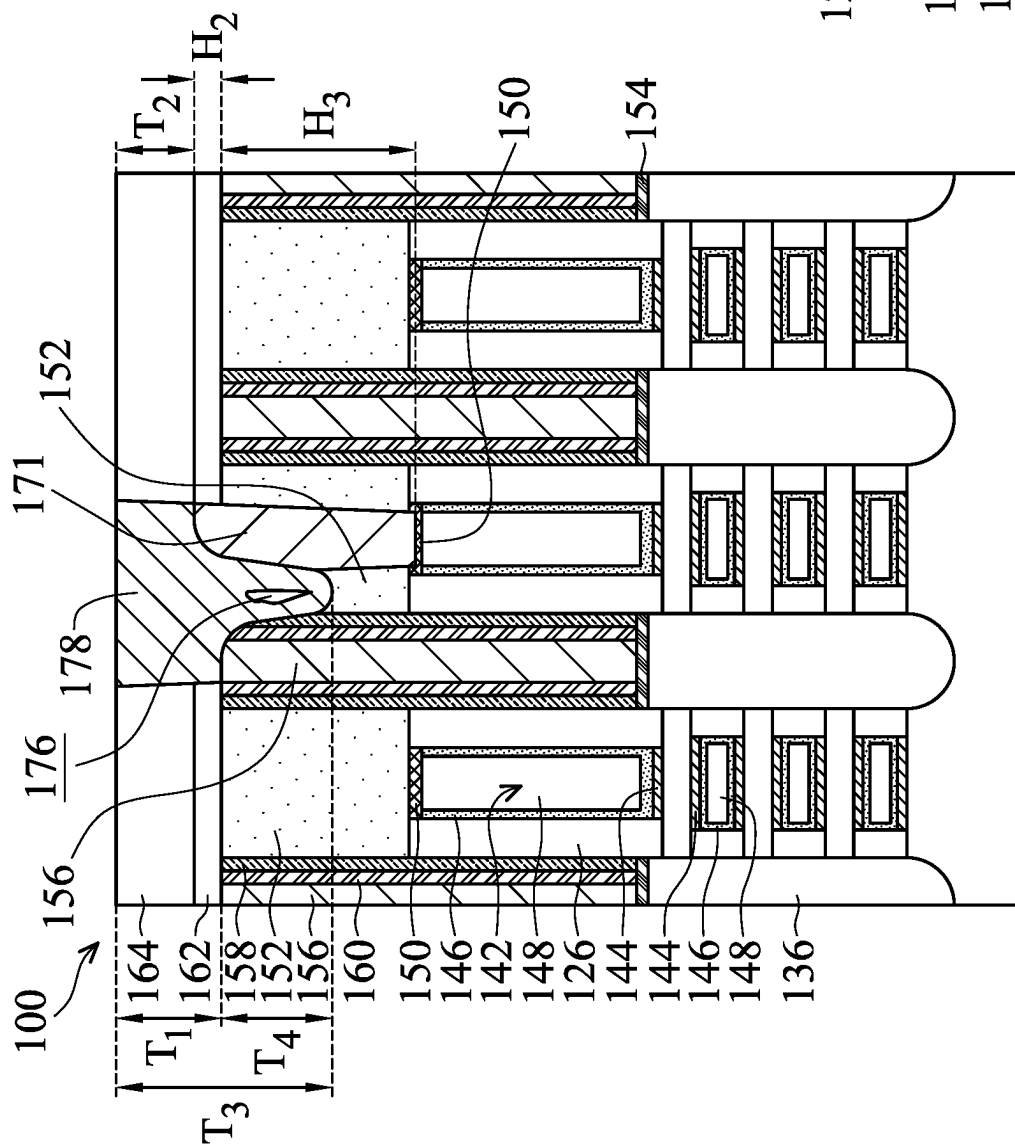


FIG. 2P-1

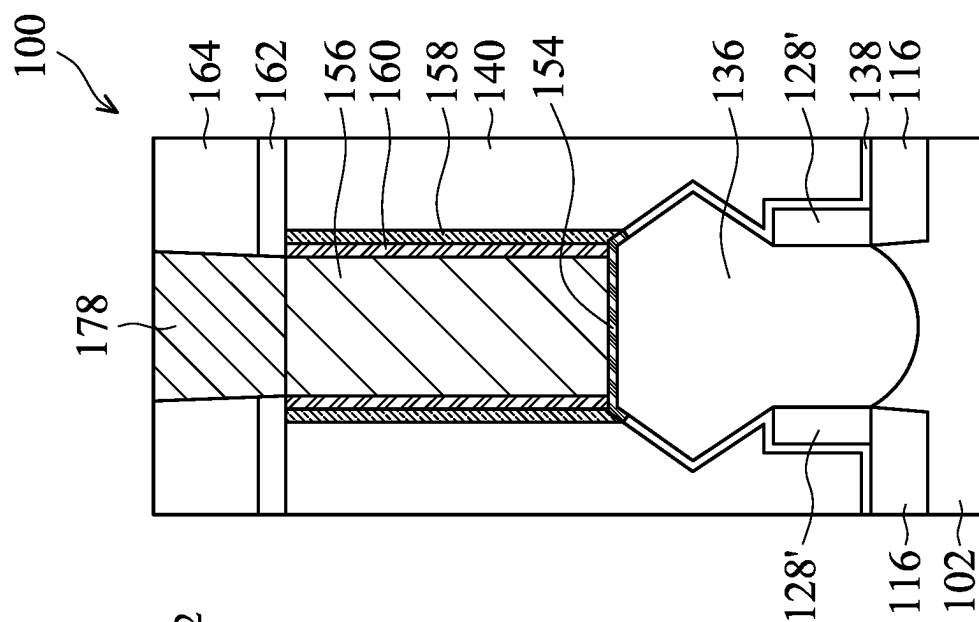


FIG. 2P-2

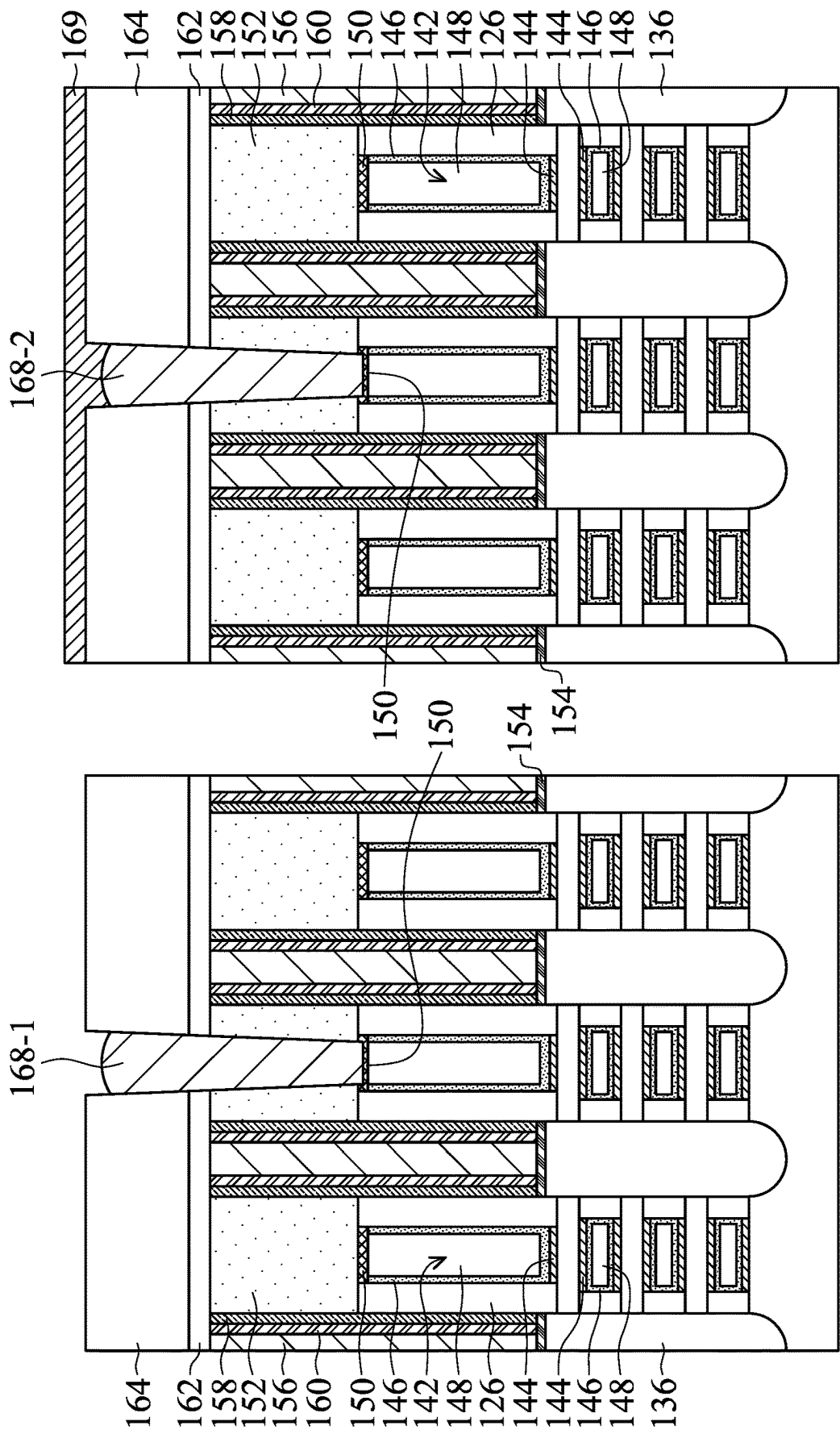


FIG. 4

FIG. 3

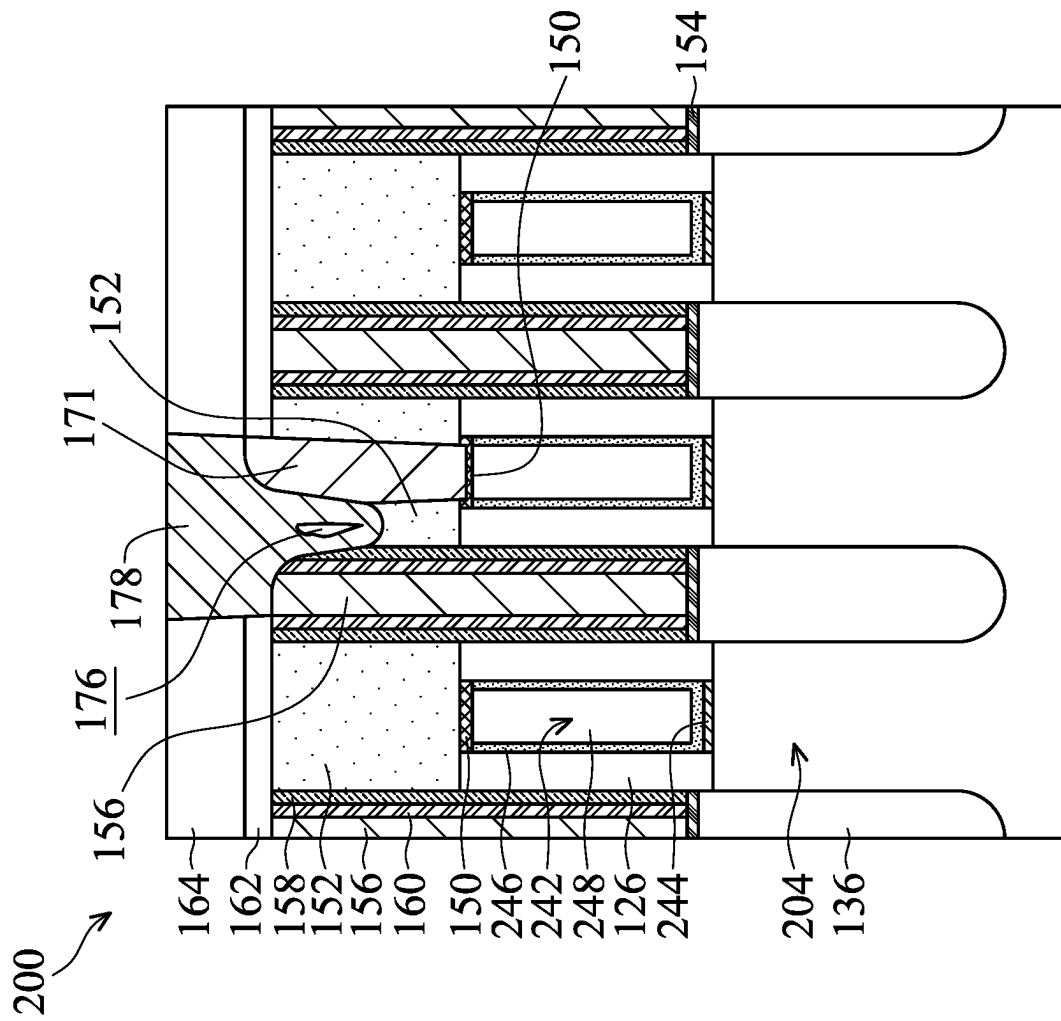


FIG. 5

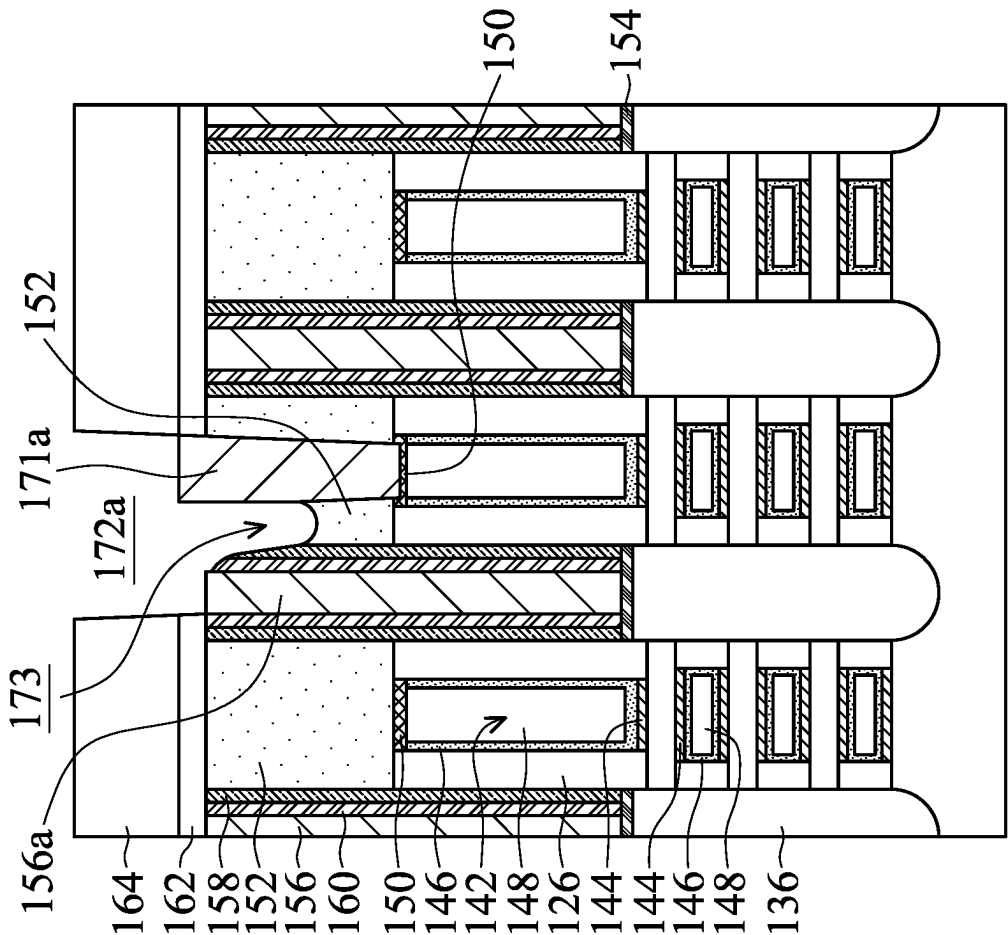


FIG. 6A-1

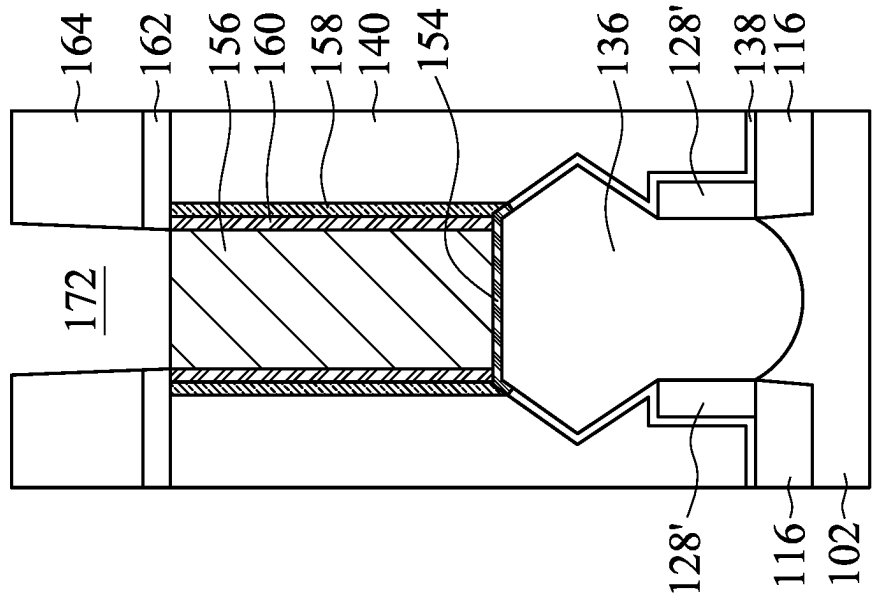


FIG. 6A-2

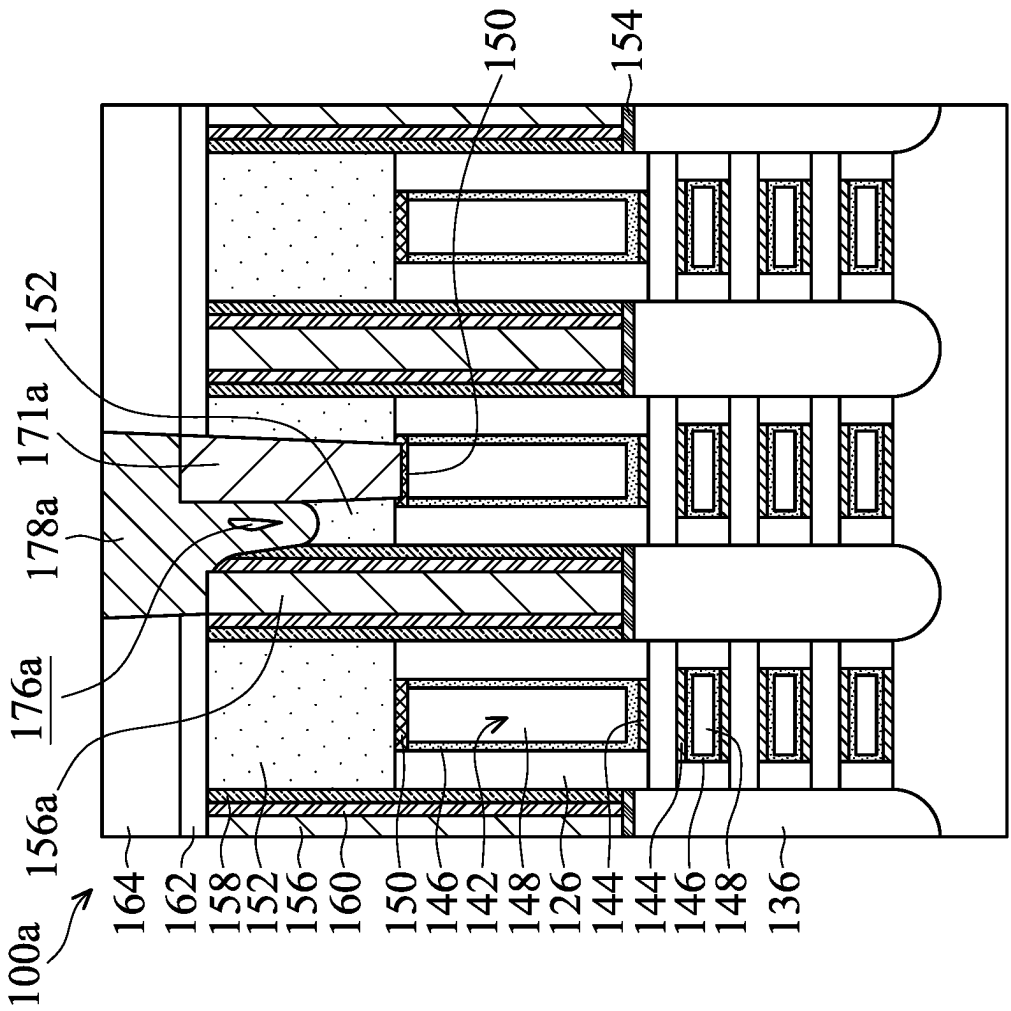


FIG. 6B-1

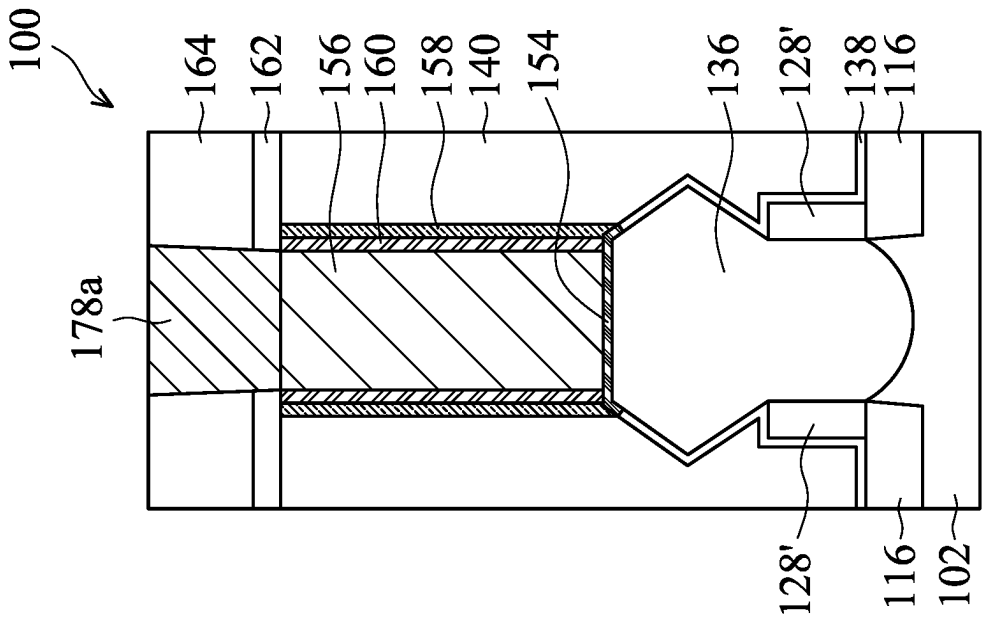


FIG. 6B-2

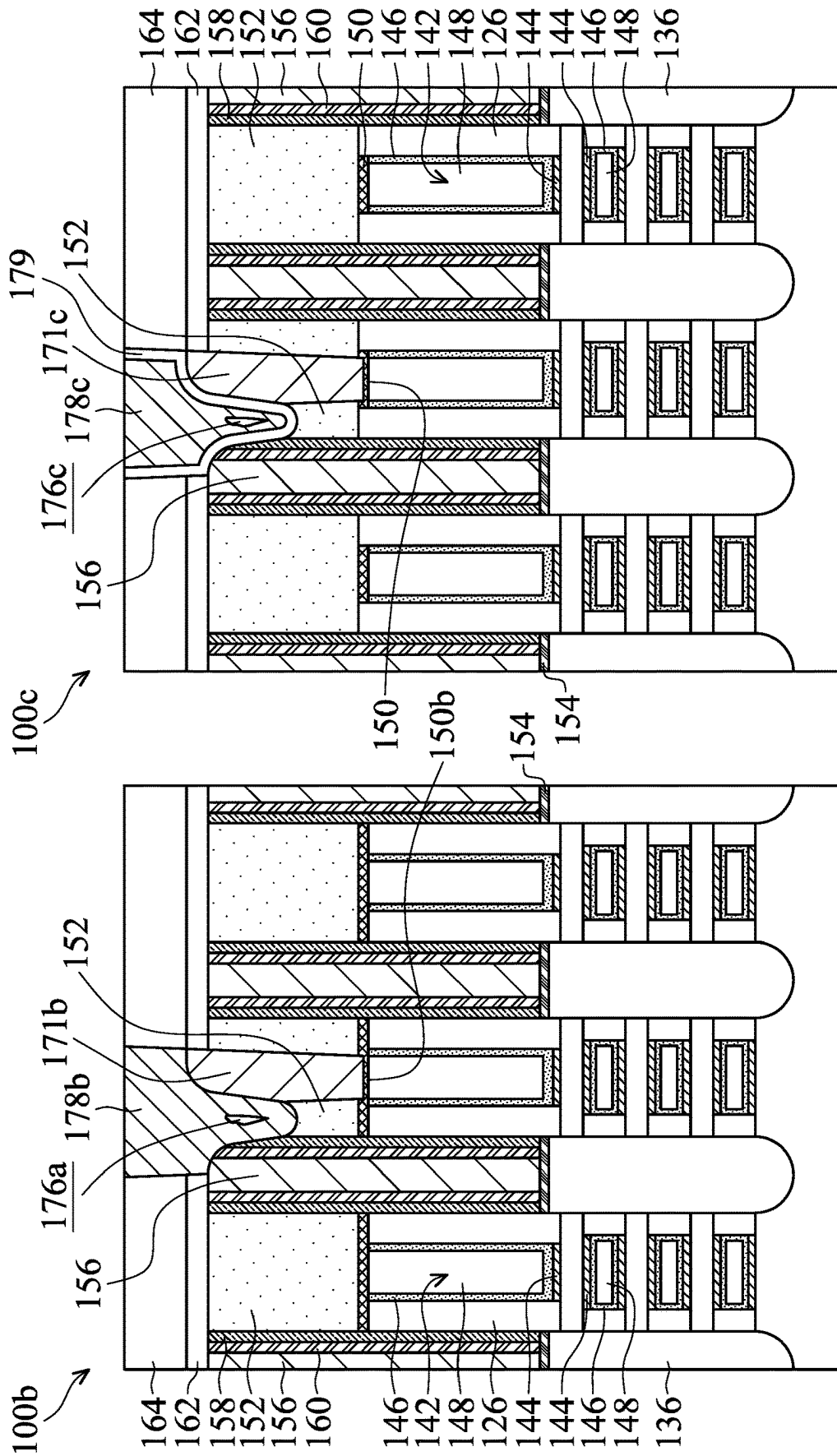


FIG. 7

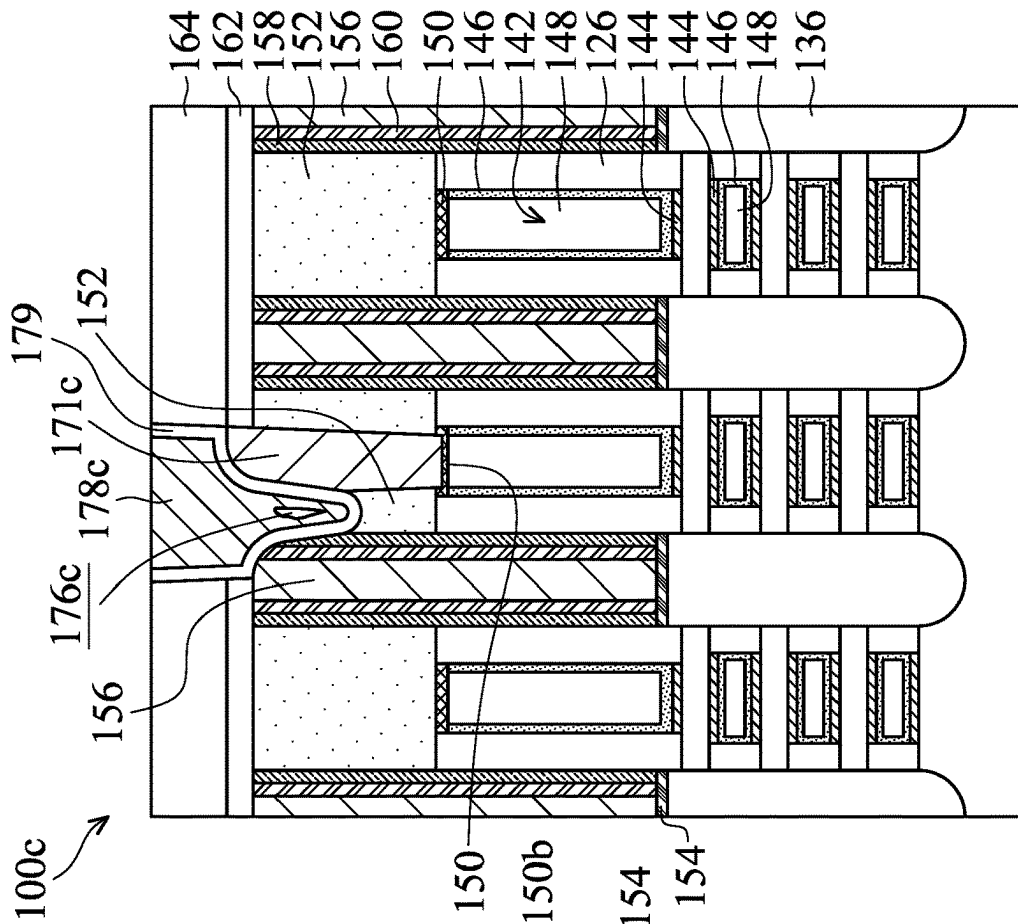


FIG. 8

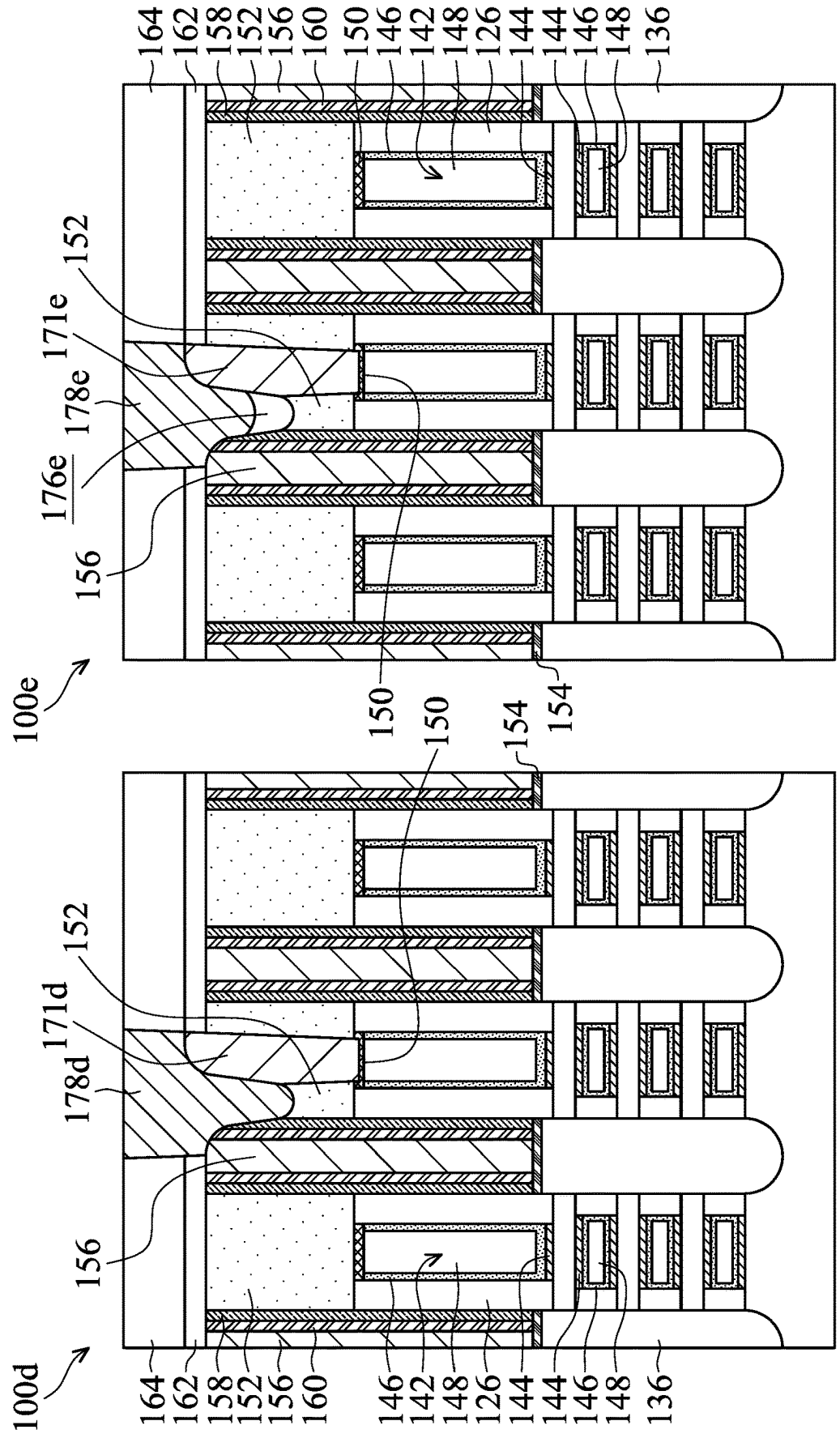


FIG. 9

FIG. 10

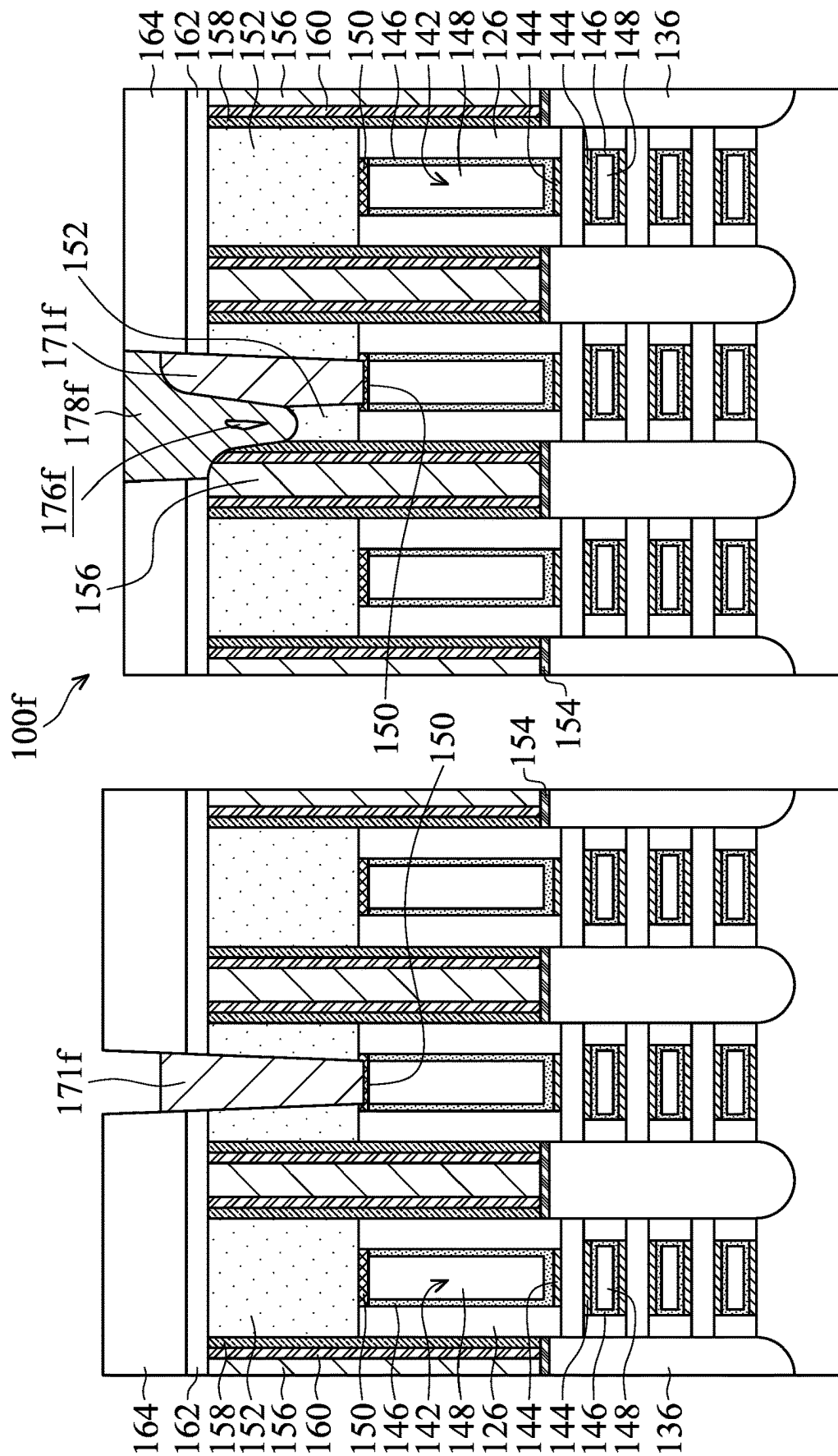


FIG. 11A

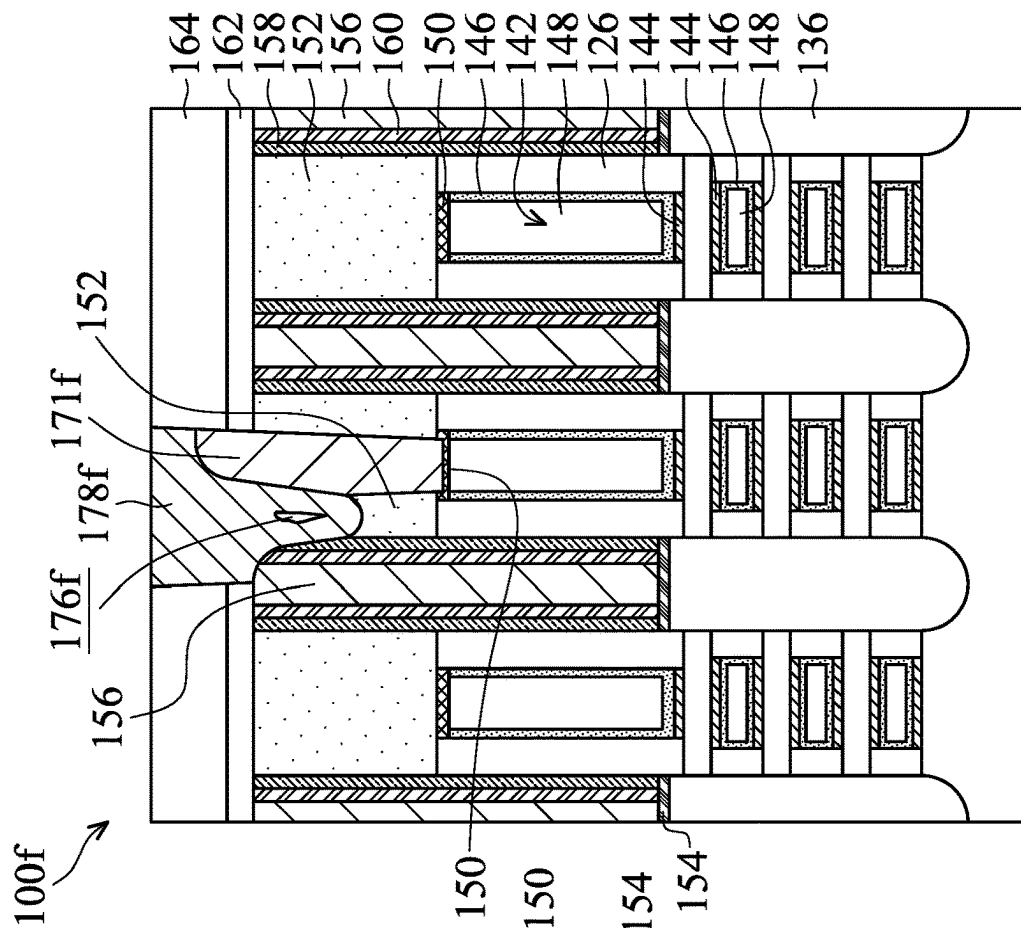


FIG. 11B

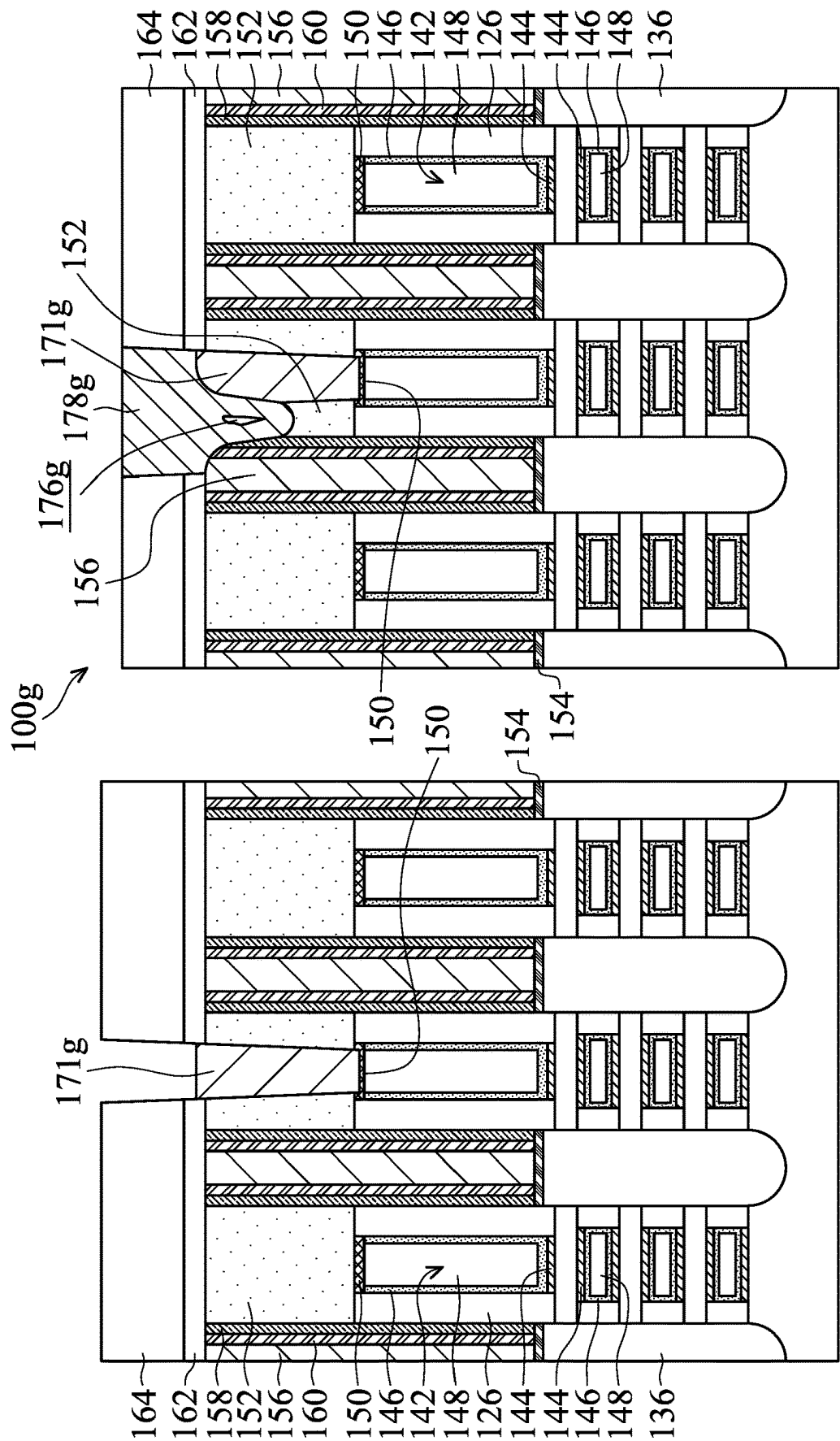


FIG. 12B

FIG. 12A

1

SEMICONDUCTOR TRANSISTOR STRUCTURE WITH NANOSTRUCTURES AND CONDUCTIVE STRUCTURE AND METHOD FOR MANUFACTURING THE SAME

PRIORITY CLAIM AND CROSS-REFERENCE

This application is a Continuation application of U.S. patent application Ser. No. 17/978,409, filed on Nov. 1, 2022, which is a Continuation application of U.S. patent application Ser. No. 17/325,419, filed on May 20, 2021, which claims the benefit of U.S. Provisional Application No. 63/158,489, filed on Mar. 9, 2021, the entirety of which is incorporated by reference herein.

BACKGROUND

The electronics industry is experiencing ever-increasing demand for smaller and faster electronic devices that are able to perform a greater number of increasingly complex and sophisticated functions. Accordingly, there is a continuing trend in the semiconductor industry to manufacture low-cost, high-performance, and low-power integrated circuits (ICs). So far, these goals have been achieved in large part by scaling down semiconductor IC dimensions (e.g., minimum feature size) and thereby improving production efficiency and lowering associated costs. However, such miniaturization has introduced greater complexity into the semiconductor manufacturing process. Thus, the realization of continued advances in semiconductor ICs and devices calls for similar advances in semiconductor manufacturing processes and technology.

Recently, multi-gate devices have been introduced in an effort to improve gate control by increasing gate-channel coupling, reduce OFF-state current, and reduce short-channel effects (SCEs). However, integration of fabrication of the multi-gate devices can be challenging.

BRIEF DESCRIPTION OF THE DRAWINGS

Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying Figures. It should be noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

FIGS. 1A, 1B, 1C, 1D, and 1E illustrate perspective views of intermediate stages of manufacturing a semiconductor structure in accordance with some embodiments.

FIGS. 2A-1, 2B-1, 2C-1, 2D-1, 2E-1, 2F-1, 2G-1, 2H-1, 2I-1, 2J-1, 2K-1, 2L-1, 2M-1, 2N-1, 2O-1, and 2P-1 illustrate cross-sectional representations of various stages of manufacturing the semiconductor structure shown along line A-A' in FIG. 1E in accordance with some embodiments.

FIGS. 2A-2, 2B-2, 2C-2, 2D-2, 2E-2, 2F-2, 2G-2, 2H-2, 2I-2, 2J-2, 2K-2, 2L-2, 2M-2, 2N-2, 2O-2, and 2P-2 illustrate cross-sectional representations of various stages of manufacturing the semiconductor structure shown along line B-B' in FIG. 1E in accordance with some embodiments.

FIG. 3 illustrates a cross-sectional view of another intermediate stage of manufacturing the semiconductor structure in accordance with some embodiments.

FIG. 4 illustrates a cross-sectional view of another intermediate stage of manufacturing the semiconductor structure in accordance with some embodiments.

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FIG. 5 illustrates a cross-sectional view of a semiconductor structure in accordance with some embodiments.

FIGS. 6A-1, 6A-2, 6B-1, and 6B-2 illustrate cross-sectional views of manufacturing a semiconductor structure in accordance with some embodiments.

FIG. 7 illustrates a cross-sectional view of a semiconductor structure in accordance with some embodiments.

FIG. 8 illustrates a cross-sectional view of a semiconductor structure in accordance with some embodiments.

FIG. 9 illustrates a cross-sectional view of a semiconductor structure in accordance with some embodiments.

FIG. 10 illustrates a cross-sectional view of a semiconductor structure 100e in accordance with some embodiments.

FIGS. 11A and 11B illustrate cross-sectional views of manufacturing a semiconductor structure in accordance with some embodiments.

FIGS. 12A and 12B illustrate cross-sectional views of manufacturing a semiconductor structure in accordance with some embodiments.

DETAILED DESCRIPTION

The following disclosure provides many different embodiments, or examples, for implementing different features of the subject matter provided. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

Some variations of the embodiments are described. Throughout the various views and illustrative embodiments, like reference numerals are used to designate like elements. It should be understood that additional operations can be provided before, during, and after the method, and some of the operations described can be replaced or eliminated for other embodiments of the method.

The gate all around (GAA) transistor structures described below may be patterned by any suitable method. For example, the structures may be patterned using one or more photolithography processes, including double-patterning or multi-patterning processes. Generally, double-patterning or multi-patterning processes combine photolithography and self-aligned processes, allowing patterns to be created that have, for example, smaller pitches than what is otherwise obtainable using a single, direct photolithography process. For example, in one embodiment, a sacrificial layer is formed over a substrate and patterned using a photolithography process. Spacers are formed alongside the patterned sacrificial layer using a self-aligned process. The sacrificial layer is then removed, and the remaining spacers may then be used to pattern the GAA structure.

The fins described below may be patterned by any suitable method. For example, the fins may be patterned using one or more photolithography processes, including double-patterning or multi-patterning processes. Generally, double-patterning or multi-patterning processes combine photolithography and self-aligned processes, allowing patterns to be created that have, for example, smaller pitches than what is otherwise obtainable using a single, direct photolithography process.

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Embodiments of semiconductor structures and methods for forming the same are provided. The semiconductor structures may include a gate structure formed over a substrate and a source/drain structure formed adjacent to the gate structure. A contact may be formed over the source/drain structure and a conductive structure may be formed to connect the contact and the gate structure. Since the contact and the gate structure may have different heights, the formation of the conductive structure may include forming a first portion over the gate structure first and then forming a second portion over the first portion and the contact. The first portion of the conductive structure formed over the gate structure may have a relatively small height difference with the contact, so that disconnection between the gate structure and the contact due to large height difference can be prevented.

FIGS. 1A to 1E illustrate perspective views of intermediate stages of manufacturing a semiconductor structure **100** in accordance with some embodiments. As shown in FIG. 1A, first semiconductor material layers **106** and second semiconductor material layers **108** are formed over a substrate **102** in accordance with some embodiments.

The substrate **102** may be a semiconductor wafer such as a silicon wafer. Alternatively or additionally, the substrate **102** may include elementary semiconductor materials, compound semiconductor materials, and/or alloy semiconductor materials. Elementary semiconductor materials may include, but are not limited to, crystal silicon, polycrystalline silicon, amorphous silicon, germanium, and/or diamond. Compound semiconductor materials may include, but are not limited to, silicon carbide, gallium arsenic, gallium phosphide, indium phosphide, indium arsenide, and/or indium antimonide. Alloy semiconductor materials may include, but are not limited to, SiGe, GaAsP, AlInAs, AlGaAs, GaInAs, GaInP, and/or GaInAsP.

In some embodiments, the first semiconductor material layers **106** and the second semiconductor material layers **108** are alternately stacked over the substrate **102**. In some embodiment, the first semiconductor material layers **106** and the second semiconductor material layers **108** are made of different semiconductor materials. In some embodiments, the first semiconductor material layers **106** are made of SiGe, and the second semiconductor material layers **108** are made of silicon. It should be noted that although three first semiconductor material layers **106** and three second semiconductor material layers **108** are formed, the semiconductor structure may include more or fewer first semiconductor material layers **106** and second semiconductor material layers **108**. For example, the semiconductor structure may include two to five of the first semiconductor material layers **106** and the second semiconductor material layers.

The first semiconductor material layers **106** and the second semiconductor material layers **108** may be formed by using low-pressure chemical vapor deposition (LPCVD), epitaxial growth process, another suitable method, or a combination thereof. In some embodiments, the epitaxial

growth process includes molecular beam epitaxy (MBE), metal organic chemical vapor deposition (MOCVD), or vapor phase epitaxy (VPE).

After the first semiconductor material layers **106** and the second semiconductor material layers **108** are formed as a semiconductor material stack over the substrate **102**, the semiconductor material stack is patterned to form a fin structure **104**, as shown in FIG. 1B in accordance with some embodiments. In some embodiments, the fin structure **104** includes a base fin structure **104B** and the semiconductor material stack of the first semiconductor material layers **106** and the second semiconductor material layers **108**.

In some embodiments, the patterning process includes forming a mask structure **110** over the semiconductor material stack, and etching the semiconductor material stack and the underlying substrate **102** through the mask structure **110**. In some embodiments, the mask structure **110** is a multilayer structure including a pad oxide layer **112** and a nitride layer **114** formed over the pad oxide layer **112**. The pad oxide layer **112** may be made of silicon oxide, which is formed by thermal oxidation or CVD, and the nitride layer **114** may be made of silicon nitride, which is formed by CVD, such as LPCVD or plasma-enhanced CVD (PECVD).

After the fin structure **104** is formed, an isolation structure **116** is formed around the fin structure **104**, and the mask structure **110** is removed, as shown in FIG. 1C in accordance with some embodiments. The isolation structure **116** is configured to electrically isolate active regions (e.g. the fin structure **104**) of the semiconductor structure **100** and is also referred to as shallow trench isolation (STI) feature in accordance with some embodiments.

The isolation structure **116** may be formed by depositing an insulating layer over the substrate **102** and recessing the insulating layer so that the fin structure **104** is protruded from the isolation structure **116**. In some embodiments, the isolation structure **116** is made of silicon oxide, silicon nitride, silicon oxynitride (SiON), another suitable insulating material, or a combination thereof. In some embodiments, a dielectric liner (not shown) is formed before the isolation structure **116** is formed, and the dielectric liner is made of silicon nitride and the isolation structure formed over the dielectric liner is made of silicon oxide.

After the isolation structure **116** is formed, dummy gate structures **118** are formed across the fin structure **104** and extend over the isolation structure **116**, as shown in FIG. 1D in accordance with some embodiments. The dummy gate structures **118** may be used to define the source/drain regions and the channel regions of the resulting semiconductor structure **100**.

In some embodiments, the dummy gate structures **118** include dummy gate dielectric layers **120** and dummy gate electrode layers **122**. In some embodiments, the dummy gate dielectric layers **120** are made of one or more dielectric materials, such as silicon oxide, silicon nitride, silicon oxynitride (SiON), HfO₂, HfZrO, HfSiO, HfTiO, HfAlO, or a combination thereof. In some embodiments, the dummy gate dielectric layers **120** are formed using thermal oxidation, CVD, ALD, physical vapor deposition (PVD), another suitable method, or a combination thereof.

In some embodiments, the conductive material includes polycrystalline-silicon (poly-Si), poly-crystalline silicon-germanium (poly-SiGe), metallic nitrides, metallic silicides, metals, or a combination thereof. In some embodiments, the dummy gate electrode layers **122** are formed using CVD, PVD, or a combination thereof.

In some embodiments, hard mask layers **124** are formed over the dummy gate structures **118**. In some embodiments,

the hard mask layers **124** include multiple layers, such as an oxide layer and a nitride layer. In some embodiments, the oxide layer is silicon oxide, and the nitride layer is silicon nitride.

The formation of the dummy gate structures **118** may include conformally forming a dielectric material as the dummy gate dielectric layers **120**. Afterwards, a conductive material may be formed over the dielectric material as the dummy gate electrode layers **122**, and the hard mask layer **124** may be formed over the conductive material. Next, the dielectric material and the conductive material may be patterned through the hard mask layer **124** to form the dummy gate structures **118**.

After the dummy gate structures **118** are formed, gate spacers **126** are formed along and covering opposite sidewalls of the dummy gate structure **118** and fin spacers **128** are formed along and covering opposite sidewalls of the source/drain regions of the fin structure **104**, as shown in FIG. 1E in accordance with some embodiments.

The gate spacers **126** may be configured to separate source/drain structures from the dummy gate structure **118** and support the dummy gate structure **118**, and the fin spacers **128** may be configured to constrain a lateral growth of subsequently formed source/drain structure and support the fin structure **104**.

In some embodiments, the gate spacers **126** and the fin spacers **128** are made of a dielectric material, such as silicon oxide (SiO₂), silicon nitride (SiN), silicon carbide (SiC), silicon oxynitride (SiON), silicon carbon nitride (SiCN), silicon oxide carbonitride (SiOCN), and/or a combination thereof. The formation of the gate spacers **126** and the fin spacers **128** may include conformally depositing a dielectric material covering the dummy gate structure **118**, the fin structure **104**, and the isolation structure **116** over the substrate **102**, and performing an anisotropic etching process, such as dry plasma etching, to remove the dielectric layer covering the top surfaces of the dummy gate structure **118**, the fin structure **104**, and portions of the isolation structure **116**.

FIGS. 2A-1 to 2P-1 illustrate cross-sectional representations of various stages of manufacturing the semiconductor structure **100** shown along line A-A' in FIG. 1E in accordance with some embodiments. FIGS. 2A-2 to 2P-2 illustrate cross-sectional representations of various stages of manufacturing the semiconductor structure **100** shown along line B-B' in FIG. 1E in accordance with some embodiments. More specifically, FIG. 2A-1 illustrates the cross-sectional representation shown along line A-A' and FIG. 2A-2 illustrates the cross-sectional representation shown along line B-B' in FIG. 1E in accordance with some embodiments.

After the gate spacers **126** and the fin spacers **128** are formed, the source/drain regions of the fin structure **104** are recessed to form source/drain recesses **130**, as shown in FIGS. 2B-1 and 2B-2 in accordance with some embodiments. More specifically, the first semiconductor material layers **106** and the second semiconductor material layers **108** not covered by the dummy gate structures **118** and the gate spacers **126** are removed in accordance with some embodiments. In addition, some portions of the base fin structure **104B** are also recessed to form curved top surfaces, as shown in FIG. 2B-1 in accordance with some embodiments.

In some embodiments, the fin structure **104** is recessed by performing an etching process. The etching process may be an anisotropic etching process, such as dry plasma etching, and the dummy gate structure **118** and the gate spacers **126** are used as etching masks during the etching process. In

some embodiments, the fin spacers **128** are also recessed to form lowered fin spacers **128'**.

After the source/drain recesses **130** are formed, the first semiconductor material layers **106** exposed by the source/drain recesses **130** are laterally recessed to form notches **132**, as shown in FIGS. 2C-1 and 2C-2 in accordance with some embodiments.

In some embodiments, an etching process is performed on the semiconductor structure **100** to laterally recess the first semiconductor material layers **106** of the fin structure **104** from the source/drain recesses **130**. In some embodiments, during the etching process, the first semiconductor material layers **106** have a greater etching rate (or etching amount) than the second semiconductor material layers **108**, thereby forming notches **132** between adjacent second semiconductor material layers **108**. In some embodiments, the etching process is an isotropic etching such as dry chemical etching, remote plasma etching, wet chemical etching, another suitable technique, and/or a combination thereof.

Next, inner spacers **134** are formed in the notches **132** between the second semiconductor material layers **108**, as shown in FIGS. 2D-1 and 2D-2 in accordance with some embodiments. The inner spacers **134** are configured to separate the source/drain structures and the gate structures formed in subsequent manufacturing processes in accordance with some embodiments. In some embodiments, the inner spacers **134** are made of a dielectric material, such as silicon oxide (SiO₂), silicon nitride (SiN), silicon carbide (SiC), silicon oxynitride (SiON), silicon carbon nitride (SiCN), silicon oxide carbonitride (SiOCN), or a combination thereof.

After the inner spacers **134** are formed, source/drain structures **136** are formed in the source/drain recesses **130**, as shown in FIGS. 2E-1 and 2E-2 in accordance with some embodiments. In some embodiments, the source/drain structures **136** are formed using an epitaxial growth process, such as MBE, MOCVD, VPE, other applicable epitaxial growth process, or a combination thereof. In some embodiments, the source/drain structures **136** are made of any applicable material, such as Ge, Si, GaAs, AlGaAs, SiGe, GaAsP, SiP, SiC, SiCP, or a combination thereof.

In some embodiments, the source/drain structures **136** are in-situ doped during the epitaxial growth process. For example, the source/drain structures **136** may be the epitaxially grown SiGe doped with boron (B). For example, the source/drain structures **136** may be the epitaxially grown Si doped with carbon to form silicon:carbon (Si:C) source/drain features, phosphorous to form silicon:phosphor (Si:P) source/drain features, or both carbon and phosphorous to form silicon carbon phosphor (SiCP) source/drain features. In some embodiments, the source/drain structures **136** are doped in one or more implantation processes after the epitaxial growth process.

After the source/drain structures **136** are formed, a contact etch stop layer (CESL) **138** is conformally formed to cover the source/drain structures **136** and an interlayer dielectric (ILD) layer **140** is formed over the contact etch stop layers **138**, as shown in FIGS. 2F-1 and 2F-2 in accordance with some embodiments.

In some embodiments, the contact etch stop layer **138** is made of a dielectric materials, such as silicon nitride, silicon oxide, silicon oxynitride, another suitable dielectric material, or a combination thereof. The dielectric material for the contact etch stop layers **138** may be conformally deposited over the semiconductor structure by performing CVD, ALD, other applicable methods, or a combination thereof.

The interlayer dielectric layer **140** may include multilayers made of multiple dielectric materials, such as silicon oxide, silicon nitride, silicon oxynitride, phosphosilicate glass (PSG), borophosphosilicate glass (BPSG), and/or other applicable low-k dielectric materials. The interlayer dielectric layer **140** may be formed by chemical vapor deposition (CVD), physical vapor deposition, (PVD), atomic layer deposition (ALD), or other applicable processes.

After the contact etch stop layer **138** and the interlayer dielectric layer **140** are deposited, a planarization process such as CMP or an etch-back process may be performed until the gate electrode layers **120** of the dummy gate structures **118** are exposed, as shown in FIG. 2F-1 in accordance with some embodiments.

Next, the dummy gate structures **118** are replaced by gate structure **142**, as shown in FIGS. 2G-1 and 2G-2 in accordance with some embodiments. More specifically, the dummy gate structures **118** and the first semiconductor material layers **106** are removed to form nanostructures **108'** with the second semiconductor material layers **108** in accordance with some embodiments. The removal process may include one or more etching processes. For example, when the dummy gate electrode layers **122** are polysilicon, a wet etchant such as a tetramethylammonium hydroxide (TMAH) solution may be used to selectively remove the dummy gate electrode layers **122**. Afterwards, the dummy gate dielectric layers **120** may be removed using a plasma dry etching, a dry chemical etching, and/or a wet etching. The first semiconductor material layers **106** may be removed by performing a selective wet etching process, such as APM (e.g., ammonia hydroxide-hydrogen peroxide-water mixture) etching process. For example, the wet etching process uses etchants such as ammonium hydroxide (NH₄OH), TMAH, ethylenediamine pyrocatechol (EDP), and/or potassium hydroxide (KOH) solutions. In some embodiments, the upper portions of the gate spacers **126** are also removed.

After the nanostructures **108'** are formed, gate structures **142** are formed wrapping around the nanostructures **108'**, as shown in FIGS. 2G-1 and 2G-2 in accordance with some embodiments. The gate structures **142** wrap around the nanostructures **108'** to form gate-all-around transistor structures in accordance with some embodiments. In some embodiments, the gate structure **142** includes an interfacial layer **144**, a gate dielectric layer **146**, and a gate electrode layer **148**.

In some embodiments, the interfacial layers **144** are oxide layers formed around the nanostructures **108'** and on the top of the base fin structure **104B**. In some embodiments, the interfacial layers **144** are formed by performing a thermal process.

In some embodiments, the gate dielectric layers **146** are formed over the interfacial layers **144**, so that the nanostructures **108'** are surrounded (e.g. wrapped) by the gate dielectric layers **146**. In addition, the gate dielectric layers **146** also cover the sidewalls of the gate spacers **126** and the inner spacers **134** in accordance with some embodiments. In some embodiments, the gate dielectric layers **146** are made of one or more layers of dielectric materials, such as HfO₂, HfSiO, HfSiON, HfTaO, HfTiO, HfZrO, zirconium oxide, aluminum oxide, titanium oxide, hafnium dioxide-alumina (HfO₂—Al₂O₃) alloy, another suitable high-k dielectric material, or a combination thereof. In some embodiments, the gate dielectric layers **146** are formed using CVD, ALD, another applicable method, or a combination thereof.

In some embodiments, the gate electrode layers **148** are formed on the gate dielectric layer **146**. In some embodi-

ments, the gate electrode layers **148** are made of one or more layers of conductive material, such as aluminum, copper, titanium, tantalum, tungsten, cobalt, molybdenum, tantalum nitride, nickel silicide, cobalt silicide, TiN, WN, TiAl, TiAlN, TaCN, TaC, TaSiN, metal alloys, another suitable material, or a combination thereof. In some embodiments, the gate electrode layers **148** are formed using CVD, ALD, electroplating, another applicable method, or a combination thereof. Other conductive layers, such as work function metal layers, may also be formed in the gate structures **142**, although they are not shown in the figures. After the interfacial layers **144**, the gate dielectric layers **146**, and the gate electrode layers **148** are formed, a planarization process such as CMP or an etch-back process may be performed until the interlayer dielectric layer **140** is exposed.

Afterwards, an etch back process is performed to remove the upper portion of the gate structures **142**, and cap layers **150** and mask layers **152** are formed over the gate structures **142**, as shown in FIGS. 2G-1 and 2G-2 in accordance with some embodiments. More specifically, the upper portion of the gate structures **142** are removed to form a recess in the interlayer dielectric layer **140** and the cap layers **150** and the mask layers **152** are formed in the recess in accordance with some embodiments.

In some embodiments, the cap layers **150** are made of W, Ti, Co, Ru, Ni, or the like. The cap layers **150** may be formed using CVD, ALD, electroplating, another applicable method, or a combination thereof. In some embodiments, the cap layers **150** are selectively formed over the gate structures **142**. In some embodiments, the mask layers **152** are made of SiO₂, Si₃N₄, SiON, SiOCN, SiOCH, or the like. The mask layers **152** may be formed using CVD, ALD, electroplating, another applicable method, or a combination thereof.

After the mask layers **152** are formed, contact openings are formed through the contact etch stop layer **138** and the interlayer dielectric layer **140**, and silicide layers **154** and contacts **156** are formed over the source/drain structures **136**, as shown in FIGS. 2H-1 and 2H-2 in accordance with some embodiments. In some embodiments, liners **158** and barrier layers **160** are formed around the contacts **156**.

More specifically, the contact openings may be formed through the contact etch stop layer **138** and the interlayer dielectric layer **140** to expose the top surfaces of the source/drain structures **136**, and the silicide layers **154** and the contacts **156** may be formed in the contact openings. The contact openings may be formed using a photolithography process and an etching process. In addition, some portions of the source/drain structures **136** exposed by the contact openings may also be etched during the etching process.

After the contact openings are formed, the silicide layers **154** may be formed by forming a metal layer over the top surface of the source/drain structures **136** and annealing the metal layer so the metal layer reacts with the source/drain structures **136** to form the silicide layers **160**. The unreacted metal layer may be removed after the silicide layers **154** are formed.

Afterwards, the liners **158**, the barrier layers **160**, and the contacts **156** are formed over the silicide layers **154** in the contact openings and a polishing process is performed, as shown in FIGS. 2H-1 and 2H-2 in accordance with some embodiments. As shown in FIG. 2H-1, the top surface of the contact **156** is substantially level with the top surface of the mask layer **152** in accordance with some embodiments.

In some embodiments, the contacts **156** are made of a conductive material including aluminum (Al), copper (Cu), tungsten (W), titanium (Ti), tantalum (Ta), titanium nitride

(TiN), cobalt, tantalum nitride (Ta₂N₃), nickel silicide (NiSi), cobalt silicide (CoSi₂), copper silicide, tantalum carbide (TaC), tantalum silicide nitride (TaSiN), tantalum carbide nitride (TaCN), titanium aluminate (TiAl), titanium aluminum nitride (TiAlN), other applicable conductive materials, or a combination thereof. In some embodiments, the liner **158** is made of silicon nitride, although any other applicable dielectric may be used as an alternative. In some embodiments, the barrier layer **160** is made of tantalum nitride, although other materials, such as tantalum, titanium, titanium nitride, or the like, may also be used. The liners **158**, the barrier layers **160**, and the contacts **156** may be formed using a process such as chemical vapor deposition (CVD), physical vapor deposition (PVD), plasma enhanced CVD (PECVD), plasma enhanced physical vapor deposition (PEPVD), atomic layer deposition (ALD), or any other applicable deposition processes.

In some embodiments, the top surface of the contact **156** is higher than the top surface of the gate structure **142** and is substantially level with the mask layer **152**. In some embodiments, the height difference between the contact **156** and the gate structure **142** is substantially level with the height of the mask layer **152**.

After the contacts **156** are formed, an etch stop layer **162** is formed over the contacts **156** and the mask layers **152**, and a dielectric layer **164** is formed over the etch stop layer **162**, as shown in FIGS. 2I-1 and 2I-2 in accordance with some embodiments.

In some embodiments, the etch stop layer **162** is made of a dielectric materials, such as silicon nitride, silicon oxide, silicon oxynitride, another suitable dielectric material, or a combination thereof. The dielectric material for the contact etch stop layers **162** may be conformally deposited over the semiconductor structure by performing CVD, ALD, other applicable methods, or a combination thereof.

The dielectric layer **164** may include multilayers made of multiple dielectric materials, such as silicon oxide, silicon nitride, silicon oxynitride, phosphosilicate glass (PSG), borophosphosilicate glass (BPSG), and/or other applicable low-k dielectric materials. The dielectric layer **164** may be formed by chemical vapor deposition (CVD), physical vapor deposition, (PVD), atomic layer deposition (ALD), or other applicable processes.

After the dielectric layer **164** is formed, a first trench **166** is formed through the mask layer **152**, the etch stop layer **162**, and the dielectric layer **164**, as shown in FIGS. 2J-1 and 2J-2 in accordance with some embodiments. In some embodiments, the upper portion of the cap layer **150** is also etched when forming the first trench **166**. The first trench **166** may be formed by performing one or more etching processes, including dry etching processes and/or wet etching processes.

Next, a conductive material **168** is formed in the first trench **166**, as shown in FIGS. 2K-1 and 2K-2 in accordance with some embodiments. In some embodiments, the conductive material **168** is W, Ru, Mo, or the like. In some embodiments, the conductive material **168** is formed by performing chemical vapor deposition (CVD), physical vapor deposition, (PVD), atomic layer deposition (ALD), or other applicable processes.

In some embodiments, the conductive material **168** is formed by performing a bottom-up deposition process. The bottom-up deposition process generally refers to a deposition process that fills an opening from bottom to top. By using the bottom-up deposition process, the first trench **166** may be filled with no air gap generated therein. In some embodiments, the bottom-up deposition process is a selec-

tive CVD process that the conductive material **168** is selectively deposited over the cap layer **150**.

In some embodiments, the conductive material **168** is Ru, and a precursor gas used in the bottom-up deposition process includes Ru(CO)₅, Ru₃(CO)₁₂, RuCl₃, Ru(od)₃, Bis(cyclopentadienyl)ruthenium(II), Ru(CO)₃C₆H₈, Ru(CO)₂(tmhd)₂, Ru(EtCp)₂, Ru(CO)₂(acac)₂, Ru(C₆H₆)(C₆H₈), Ru(DMBD)(CO)₃, a combination thereof, or the like. In some embodiments, the conductive material **168** is W, and a precursor gas used in the bottom-up deposition process includes W(CO)₆, W(F)₆, or the like. In some embodiments, the conductive material **168** is Mo, and a precursor gas used in the bottom-up deposition process includes MoF₆, Mo(CO)₆, MoCl₅, MoO₃Cl₂, or the like.

Afterward, a polishing process is performed to form a conductive structure **170** in the first trench **166** over the gate structure **142**, as shown in FIGS. 2L-1 and 2L-2 in accordance with some embodiments. In some embodiments, a CMP process is performed to polish the conductive material **168** so that the top surface of the conductive structure **170** is substantially level with the top surface of the dielectric layer **164**.

Next, the conductive structure **170** is shortened to form a shortened conductive structure **171**, as shown in FIGS. 2M-1 and 2M-2 in accordance with some embodiments. More specifically, the upper portion of the conductive structure **170** is etched back, so that the top surface of the shortened conductive structure **171** is substantially level with the top surface of the etch stop layer **162** in accordance with some embodiments.

In some embodiments, the height H₁ of the shortened conductive structure **171** is in a range from about 10 nm to about 50 nm. In some embodiments, the depth D₁ of the trench **167** (e.g. the height of the removing upper portion of the conductive structure **170**) is in a range from about 30 nm to about 70 nm. In some embodiments, the ratio of the height H₁ to the depth D₁ is in a range from about 0.6 to about 7.

After the conductive structure **170** is shortened, a second trench **172** is formed through the dielectric layer **164** and the etch stop layer **162** to expose both the top surface of the contact **156** and the top surface of the shortened conductive structure **171**, as shown in FIGS. 2N-1 and 2N-2 in accordance with some embodiments. The second trench **172** may be formed by performing an etching process. During the etching process, the corner of the contact **156** and the corner of the shortened conductive structure **171** are both etched, such that the contact **156** and the shortened conductive structure **171** have rounded corners facing each other in accordance with some embodiments.

In some embodiments, a portion of the mask layer **152** is also removed, such that the second trench **172** has an extending portion **173** extending into the mask layer **142**. In some embodiments, the bottommost portion of the extending portion **173** of the second trench **172** is lower than the top surface of the contact **156**. In addition, an upper portion of the sidewall of the shortened conductive structure **171** is exposed by the extending portion **173** of the second trench **172** in accordance with some embodiments.

Next, a conductive material **174** is formed in the second trench **172**, as shown in FIGS. 2O-1 and 2O-2 in accordance with some embodiments. Since the corner of the shortened conductive structure **171** is etched, the shortened conductive structure **171** and the conductive material **174** have a curve interface in accordance with some embodiments.

In some embodiments, the conductive material **174** is W, Ru, Mo, Cu, or the like. In some embodiments, the conductive material **174** and the conductive material **168** are the

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same metal. In some embodiments, the conductive material **174** and the conductive material **168** are different metals.

In some embodiments, the conductive material **174** is formed by performing chemical vapor deposition (CVD), physical vapor deposition, (PVD), atomic layer deposition (ALD), or other applicable processes. In some embodiments, the conductive material **174** is formed by performing a bottom-up deposition process. The bottom-up deposition process used for forming the conductive material **174** may be the same as, or similar to, those for forming the conductive material **168** and are not repeated herein. In some embodiments, the deposition processes (e.g. precursor gas) for the conductive material **174** and the conductive material **168** are the same.

Since the shortened conductive structure **171** is formed first, the height difference of different regions of the second trench **172** (e.g. the region over the contact **156** and the region over the gate structure **142**) can be relatively small. Accordingly, the conductive material **174** can be formed by performing a bottom-up deposition process with improved connection between the contact **156** and the gate structure **142**.

In some embodiments, a void **176** is formed in the conductive material **174** at the extending portion **173** of the second trench **172**. In some embodiments, the void **176** is sandwiched between the contact **156** and the shortened conductive structure **171** and is directly over the mask layer **152**.

Afterward, a polishing process is performed to form a conductive structure **178** in the semiconductor structure **100**, as shown in FIGS. 2P-1 and 2P-2 in accordance with some embodiments. In some embodiments, a CMP process is performed to polish the conductive material **174**, so that the top surface of the conductive structure **178** is substantially level with the top surface of the dielectric layer **164**.

In some embodiments, the height difference H_2 between the shortened conductive structure **171** and the contact **156** is smaller than 10 nm. Since the height difference H_2 between the shortened conductive structure **171** and the contact **156** is relatively small, the conductive structure **178** may be better formed. That is, the risk of disconnection between the gate structure **142** and the contact **156** due to the height difference between the gate structure **142** and the contact **156** can be reduced. In some embodiments, the ratio of the height difference H_2 between the shortened conductive structure **171** and the contact **156** to the height difference H_3 of the contact **156** and the gate structure **142** is in a range from about 0.2 to about 0.5.

As shown in FIG. 2P-1, the conductive structure **178** may have different thicknesses at different portions. In some embodiments, the conductive structure **178** has a first portion over the contact **156**, and the thickness T_1 of the first portion is in a range from about 35 nm to about 90 nm. In some embodiments, the conductive structure **178** has a second portion over the shortened conductive structure **171**, and the thickness T_2 of the second portion is in a range from about 30 nm to about 70 nm. In some embodiments, the ratio of the thickness T_2 to the thickness T_1 is in a range from about 0.5 to about 3.

In some embodiments, the conductive structure **178** has a third portion between the first portion and the second portion, and the thickness T_3 of the third portion is in a range from about 38 nm to about 100 nm. More specifically, the conductive structure **178** has an extending portion interposing between the shortened conductive structure **171** and the contact **156**, and the thickness T_4 of the extending portion is in a range from about 3 nm to about 10 nm.

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In some embodiments, the bottommost portion of the conductive structure **178** (e.g. the bottommost portion of the extending portion of the conductive structure **178**) is lower than the topmost portions (e.g. top surfaces) of the shortened conductive structure **171**, the contact **156**, the mask layer **152**, and the etch stop layer **162**. In addition, the bottommost portion of the conductive structure **178** is higher than the bottommost surfaces of the shortened conductive structure **171**, the contact **156**, and the mask layer **152**.

It is appreciated that although the cross-sectional views shown in FIGS. 2P-1 and 2P-2 are described with reference to a method, the structures are not limited to the method but rather may stand alone separate of the method.

FIG. 3 illustrates a cross-sectional view of another intermediate stage of manufacturing the semiconductor structure **100** in accordance with some embodiments. The processes for manufacturing the semiconductor structure **100** described above may be performed, except a conductive material **168-1** formed in the first trench may be thinner than the conductive material **168** shown in FIG. 2K-1 in accordance with some embodiments.

More specifically, the processes shown in FIGS. 1A to 1E, 2A-1 to 2J-1, and 2A-2 to 2J-2 are performed to form a first trench (e.g. the first trench **166**) through the mask layer **152**, the etch stop layer **162**, and the dielectric layer **164** in accordance with some embodiments. Afterwards, a conductive material **168-1** is formed in the first trench, as shown in FIG. 3 in accordance with some embodiments. The conductive material **168-1** may be similar to the conductive material **168**, except the conductive material **168-1** does not cover the top surface of the dielectric layer **164**. The processes and materials for forming the conductive material **168-1** may be similar to, or the same as, those for forming the conductive material **168** and are not repeated herein.

In some embodiments, the top surface of the conductive material **168-1** is lower than the top surface of the dielectric layer **164**, such that the first trench is not fully filled with the conductive material **168-1**. After the conductive material **168-1** is formed, a polishing process is performed to form the conductive structure over the gate structure **142** in accordance with some embodiments. During the polishing process, the upper portion of the conductive material **168-1** and the upper portion of the dielectric layer **164** are removed, so that the resulting conductive structure and the dielectric layer **164** can still have substantially flat top surfaces (similar to that shown in FIGS. 2L-1 and 2L-2) in accordance with some embodiments. Afterwards, the processes shown in FIGS. 2M-1 to 2P-1 and 2M-2 to 2P-2 may be performed to form the semiconductor structure **100**.

FIG. 4 illustrates a cross-sectional view of another intermediate stage of manufacturing the semiconductor structure **100** in accordance with some embodiments. The processes for manufacturing the semiconductor structure **100** described above may be performed, except a conductive material **168-2** and an addition conductive material **169** are formed in accordance with some embodiments.

More specifically, the processes shown in FIGS. 1A to 1E, 2A-1 to 2J-1, and 2A-2 to 2J-2 are performed to form a first trench (e.g. the first trench **166**) through the mask layer **152**, the etch stop layer **162**, and the dielectric layer **164** in accordance with some embodiments. Afterwards, the conductive material **168-2** is formed in the first trench, and the conductive material **169** is formed over the conductive material **168-2**, as shown in FIG. 4 in accordance with some embodiments. The processes and materials for forming the

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conductive material **168-2** may be similar to, or the same as, those for forming the conductive material **168** and are not repeated herein.

Similar to the conductive material **168-1**, the top surface of the conductive material **168-2** is lower than the top surface of the dielectric layer **164**, such that the first trench is not fully filled with the conductive material **168-2** in accordance with some embodiments. In addition, the upper portion of the first trench is filled with the conductive material **169** and the top surface of the dielectric layer **164** is covered by the conductive material **169** in accordance with some embodiments.

In some embodiments, the conductive material **168-2** and the conductive material **169** are made of the same material but are formed by different processes. In some embodiments, the conductive material **169** is formed by performing a CVD process.

After the conductive material **169** is formed, a polishing process is performed to form the conductive structure over the gate structure **142** in accordance with some embodiments. In some embodiments, the conductive material **169** are completely removed during the polishing process, so that the resulting conductive structure is completely made of the conductive material **168-2**. In some embodiments, the upper portion of the conductive material **168-2** and the upper portion of the dielectric layer **164** are also removed during the polishing process. Since the upper portion of the first trench is filled with the conductive material **169** and the top surface of the dielectric layer **164** is covered by the conductive material **169**, the uniformity of the polishing process may be improved. Afterwards, the processes shown in FIGS. **2M-1** to **2P-1** and **2M-2** to **2P-2** may be performed to form the semiconductor structure **100**.

FIG. **5** illustrates a cross-sectional view of a semiconductor structure **200** in accordance with some embodiments. The semiconductor structure **200** may be similar to the semiconductor structure **100** shown in FIGS. **2P-1** and **2P-2**, except the semiconductor structure **200** is a FinFET structure in accordance with some embodiments. Materials and processes for manufacturing the semiconductor structure **200** may be similar to, or the same as, those for manufacturing the semiconductor structure **100** described above and are not repeated herein.

More specifically, a fin structure **204** is formed protruding from the substrate **102** and a gate structure **242** is formed across the fin structure **204**, as shown in FIG. **5** in accordance with some embodiments. The gate structure **242** includes an interfacial layer **244**, a gate dielectric layer **246**, and a gate electrode layer **248** in accordance with some embodiments. The processes and materials for forming the interfacial layer **244**, the gate dielectric layer **246**, and the gate electrode layer **248** are the same as those for forming the interfacial layer **144**, the gate dielectric layer **146**, and the gate electrode layer **148** and are not repeated herein. Similar to those shown in FIGS. **2P-1** and **2P-2**, the shortened conductive structure **171** is formed over the gate structure **242** and the conductive structure **178** covers both the contact **156** and the shortened conductive structure **171** over the gate structure **242**, as shown in FIG. **5** in accordance with some embodiments.

FIGS. **6A-1**, **6A-2**, **6B-1**, and **6B-2** illustrate cross-sectional views of manufacturing a semiconductor structure **100a** in accordance with some embodiments. The semiconductor structure **100a** may be the same as the semiconductor structure **100**, except the shapes of the contact and the shortened conductive structure are different in accordance with some embodiments. Materials and processes for manu-

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facturing the semiconductor structure **100a** may be similar to, or the same as, those for manufacturing the semiconductor structure **100** described above and are not repeated herein.

More specifically, FIGS. **6A-1** and **6B-1** are cross-sectional views shown along the fin direction and FIGS. **6A-2** and **6B-2** are cross-sectional views shown along the gate direction. Similar to those for forming the semiconductor structure **100**, the processes shown in FIGS. **1A** to **1E**, **2A-1** to **2M-1**, and **2A-2** to **2M-2** are performed to form a shortened conductive structure **171a** in accordance with some embodiments. Afterwards, the etching process is performed to form a second trench **172a** through the dielectric layer **164** and the etch stop layer **162**, as shown in FIGS. **6A-1** and **6B-1** in accordance with some embodiments. During the etching process, the etchant may be chosen to have a good etching selectivity between the conductive materials and the dielectric materials, so that the shapes of the contact **156a** and the shortened conductive structure **171a** may substantially remain.

Next, the processes shown in FIGS. **2O-1**, **2O-2**, **2P-1**, and **2P-2** and described previously are performed to form a conductive structure **178a** in the semiconductor structure **100a**, as shown in FIGS. **6B-1** and **6B-2** in accordance with some embodiments. In addition, a void **176a** is formed in the conductive structure **178a** and is also sandwiched between the shortened conductive structure **171a** and the contact **156a** in accordance with some embodiments. The processes and materials for forming the contact **156a**, the shortened conductive structure **171a**, and the conductive structure **178a** may be the same as those for forming the contact **156**, the shortened conductive structure **171**, and the conductive structure **178** described above and are not repeated herein.

FIG. **7** illustrates a cross-sectional view of a semiconductor structure **100b** in accordance with some embodiments. The semiconductor structure **100b** may be the same as the semiconductor structure **100** shown in FIGS. **2P-1** and **2P-2**, except its cap layer **150b** extends over the gate spacers **126** in accordance with some embodiments. Materials and processes for manufacturing the semiconductor structure **100b** may be similar to, or the same as, those for manufacturing the semiconductor structures **100** described above and are not repeated herein.

More specifically, the cap layer **150b** is sandwiched between the shortened conductive structure **171b** and the gate structure **142** and is also sandwiched between the mask layer **152** and the gate spacer **126** in accordance with some embodiments. In some embodiments, the cap layer **150b** is in direct contact with the liner **158** around the contacts **156**. In some embodiments, the cap layer **150b** vertically overlaps the void **176b** in the conductive structure **178b**.

In some embodiments, the cap layers **150b** are made of W, Ti, Co, Ru, Ni, or the like. The cap layers **150b** may be formed using CVD, ALD, electroplating, another applicable method, or a combination thereof. The processes and materials for forming the shortened conductive structure **171b** and the conductive structure **178b** may be the same as those for forming the shortened conductive structure **171** and the conductive structure **178** described above and are not repeated herein.

FIG. **8** illustrates a cross-sectional view of a semiconductor structure **100c** in accordance with some embodiments. The semiconductor structure **100c** may be the same as the semiconductor structure **100** shown in FIGS. **2P-1** and **2P-2**, except a barrier layer **179** is formed around a conductive structure **178c** in accordance with some embodiments. Materials and processes for manufacturing the semiconduc-

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tor structure **100c** may be similar to, or the same as, those for manufacturing the semiconductor structures **100** described above and are not repeated herein.

More specifically, after the second trench is formed (e.g. the second trench **172** as shown in FIGS. 2N-1 and 2N-2), the barrier layer **179** is formed lining the second trench and the conductive structure **178c** is formed over the barrier layer **179**, as shown in FIG. 8 in accordance with some embodiments. In some embodiments, the barrier layer **179** is made of tantalum nitride, although other materials, such as tantalum, titanium, titanium nitride, or the like, may also be used. The barrier layers **179** may be formed by using a process such as chemical vapor deposition (CVD), physical vapor deposition (PVD), plasma enhanced CVD (PECVD), plasma enhanced physical vapor deposition (PEPVD), atomic layer deposition (ALD), or any other applicable deposition processes.

In some embodiments, a portion of the barrier layer **179** is sandwiched between the shortened conductive structure **171c** and the contact **156**. In some embodiments, the barrier layer **179** is in direct contact with the contact **156**, the mask layer **152**, and the shortened conductive structure **171c**. In some embodiments, the bottommost portion of the barrier layer **179** is lower than a topmost portion of the mask layer **152**, the topmost portion of the contact **156**, and the topmost portion of the shortened conductive structure **171c**. The processes and materials for forming the shortened conductive structure **171c** and the conductive structure **178c** may be the same as those for forming the shortened conductive structure **171** and the conductive structure **178** described above and are not repeated herein.

FIG. 9 illustrates a cross-sectional view of a semiconductor structure **100d** in accordance with some embodiments. The semiconductor structure **100d** may be the same as the semiconductor structure **100** shown in FIGS. 2P-1 and 2P-2, except no void is formed in a conductive structure **178d** in accordance with some embodiments. Materials and processes for manufacturing the semiconductor structure **100d** may be similar to, or the same as, those for manufacturing the semiconductor structures **100** described above and are not repeated herein.

More specifically, after the second trench is formed (e.g. the second trench **171** shown in FIGS. 2N-1 and 2N-2), the second trench is completely filled with the conductive structure **178d**, as shown in FIG. 9 in accordance with some embodiments. The processes and materials for forming the shortened conductive structure **171d** and the conductive structure **178d** may be the same as those for forming the shortened conductive structure **171** and the conductive structure **178** described above and are not repeated herein.

FIG. 10 illustrates a cross-sectional view of a semiconductor structure **100e** in accordance with some embodiments. The semiconductor structure **100e** may be the same as the semiconductor structure **100** shown in FIGS. 2P-1 and 2P-2, except a void **176e** is formed between a conductive structure **178e** and a mask layer **152** in accordance with some embodiments. Materials and processes for manufacturing the semiconductor structure **100e** may be similar to, or the same as, those for manufacturing the semiconductor structures **100** described above and are not repeated herein.

More specifically, after the second trench is formed, the bottom portion of the extending portion (e.g. the extending portion **173** shown in FIG. 2N-1) of the second trench is not filled with the conductive structure **178e**, such that the void **176e** is formed, as shown in FIG. 10 in accordance with some embodiments. In some embodiments, a sidewall of the shortened conductive structure **171e** and the top surface of

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the mask layer **152** are exposed by the void **176e**. In some embodiments, the bottommost region of the void **176e** is lower than the bottommost portion of the conductive structure **178e**. The processes and materials for forming the shortened conductive structure **171e** and the conductive structure **178e** may be the same as those for forming the shortened conductive structure **171** and the conductive structure **178** described above and are not repeated herein.

FIGS. 11A and 11B illustrate cross-sectional views of manufacturing a semiconductor structure **100f** in accordance with some embodiments. The semiconductor structure **100f** may be the same as the semiconductor structure **100**, except its shortened conductive structure **171f** is higher than the shortened conductive structure **171** in accordance with some embodiments. Materials and processes for manufacturing the semiconductor structure **100f** may be similar to, or the same as, those for manufacturing the semiconductor structure **100** described above and are not repeated herein.

More specifically, the processes shown in FIGS. 1A to 1E, 2A-1 to 2L-1, and 2A-2 to 2L-2 may be performed to form a conductive structure (e.g. the conductive structure **170** shown in FIG. 2L-1). Next, the conductive structure is shortened to form a shortened conductive structure **171f**, and the top surface of the shortened conductive structure **171f** is higher than the top surface of the etch stop layer **162**, as shown in FIG. 11A in accordance with some embodiments.

Next, the processes shown in FIGS. 2N-1 to 2P-1 and 2N-2 to 2P-2 and described previously are performed to form a conductive structure **178f** in the semiconductor structure **100f**, as shown in FIG. 11B in accordance with some embodiments. In addition, a void **176f** is formed in the conductive structure **178f** and is sandwiched between the shortened conductive structure **171f** and the contact **156** in accordance with some embodiments. Since the sidewall of the etch stop layer **162** is completely covered by the shortened conductive structure **171f**, the etch stop layer **162** can be protected during the etching process for forming the second trench, and therefore the risk of short circuit between the conductive structure **178f** and the neighboring contact **156** (e.g. the contact in the right shown in FIG. 11B) can be reduced.

The processes and materials for forming the shortened conductive structure **171f** and the conductive structure **178f** may be the same as those for forming the shortened conductive structure **171** and the conductive structure **178** described above and are not repeated herein.

FIGS. 12A and 12B illustrate cross-sectional views of manufacturing a semiconductor structure **100g** in accordance with some embodiments. The semiconductor structure **100g** may be the same as the semiconductor structure **100**, except its shortened conductive structure **171g** is lower than the shortened conductive structure **171** in accordance with some embodiments. Materials and processes for manufacturing the semiconductor structure **100g** may be similar to, or the same as, those for manufacturing the semiconductor structure **100** described above and are not repeated herein.

More specifically, the processes shown in FIGS. 1A to 1E, 2A-1 to 2L-1, and 2A-2 to 2L-2 may be performed to form a conductive structure (e.g. the conductive structure **170** shown in FIG. 2L-1). Next, the conductive structure is shortened to form a shortened conductive structure **171g**, and the top surface of the shortened conductive structure **171g** is lower than the top surface of the etch stop layer **162** as shown in FIG. 12A in accordance with some embodiments.

Next, the processes shown in FIGS. 2N-1 to 2P-1 and 2N-2 to 2P-2 and described previously are performed to form a conductive structure **178g** in the semiconductor

structure **100g**, as shown in FIG. **12B** in accordance with some embodiments. In addition, a void **176g** is formed in the conductive structure **178g** and is also sandwiched between the shortened conductive structure **171g** and the contact **156** in accordance with some embodiments. Since the sidewall of the etch stop layer **162** is partially covered by the shortened conductive structure **171g**, the etch stop layer **162** can still be protected during the etching process and the lateral etching of the etch stop layer **162** can be reduced.

The processes and materials for forming the shortened conductive structure **171g** and the conductive structure **178g** may be the same as those for forming the shortened conductive structure **171** and the conductive structure **178** described above and are not repeated herein.

It should be appreciated that the semiconductor structures **100a** to **100g** having the shortened conductive structure **171a** to **171g** and the conductive structure **178a** to **178g** described above may also be applied to FinFET structures, similar to that shown in FIG. **5**, although not shown in the figures.

Generally, a conductive structure may be formed to connect a gate structure and a contact over S/D structure. However, as the device size shrank down, the conductive structure may need to be formed in a relatively small trench. In some cases, a bottom-up deposition may be performed to fill the conductive material in the small trench, so that less air gaps will be formed in the trench. However, since the contact may be much higher than the gate structure, when the conductive materials formed in the trench, the conductive material formed over the contact may block the opening of the trench first, such that the conductive material over the gate structure (i.e. at deeper portion of the trench) may not be completely filled yet. That is, the contact and the gate structure may not be well-connected.

Accordingly, a first conductive structure (e.g. the conductive structure **170**) is formed over the gate structure **142** first, and an upper portion of the conductive structure is removed to form a shortened conductive structure (e.g. the shortened conductive structure **171** and **171a** to **171g**) in accordance with some embodiments. By forming the shortened conductive structure, the second conductive structure (e.g. the conductive structure **178** and **178a** to **178g**) formed afterwards can be formed by a bottom-up deposition without worrying the blocking issues due to the relatively great height difference described above. In addition, since the shortened conductive structure and the second conductive structure formed afterwards may both be formed by performing bottom-up depositions, the filling of the trench may be improved and the performance of the resulting semiconductor structure (e.g. the semiconductor structure **100**, **100a** to **100g**, and **200**) may therefore also be improved.

It should be noted that same elements in FIGS. **1A** to **12B** may be designated by the same numerals and may include similar or the same materials and may be formed by similar or the same processes; therefore such redundant details are omitted in the interest of brevity. In addition, although FIGS. **1A** to **12B** are described in relation to the method, it will be appreciated that the structures disclosed in FIGS. **1A** to **12B** are not limited to the method but may stand alone as structures independent of the method. Similarly, although the methods shown in FIGS. **1A** to **12B** are not limited to the disclosed structures but may stand alone independent of the structures. Furthermore, the nanostructures described above may include nanowires, nanosheets, or other applicable nanostructures in accordance with some embodiments.

Also, while disclosed methods are illustrated and described below as a series of acts or events, it will be

appreciated that the illustrated ordering of such acts or events may be altered in some other embodiments. For example, some acts may occur in different orders and/or concurrently with other acts or events apart from those illustrated and/or described above. In addition, not all illustrated acts may be required to implement one or more aspects or embodiments of the description above. Further, one or more of the acts depicted above may be carried out in one or more separate acts and/or phases.

Furthermore, the terms “approximately,” “substantially,” “substantial” and “about” describe above account for small variations and may be varied in different technologies and be in the deviation range understood by the skilled in the art. For example, when used in conjunction with an event or circumstance, the terms can refer to instances in which the event or circumstance occurs precisely as well as instances in which the event or circumstance occurs to a close approximation.

Embodiments for forming semiconductor structures may be provided. The semiconductor structure may include forming a gate structure, a source/drain structure adjacent to the gate structure, and a contact over the source/drain structure. A first conductive structure may be formed over the gate structure, and the upper portion of the first conductive structure may be removed afterwards. A second conductive structure may be formed over the contact and the first conductive structure. Since the height difference between the shortened first conductive structure and the contact is relatively small, the second conductive structure may be better formed and the performance of the semiconductor structure may be improved.

In some embodiments, a semiconductor structure is provided. The semiconductor structure includes nanostructures spaced apart from each other in a first direction and a gate structure formed over and around the nanostructures. The semiconductor structure further includes a gate spacer covering a sidewall of the gate structure and a source/drain structure attached to the nanostructures in a second direction. The semiconductor structure further includes a contact spaced apart from the gate structure by the gate spacer in the second direction and a first conductive structure landing over the gate structure. The semiconductor structure further includes a second conductive structure formed over the gate spacer. In addition, a portion of the second conductive structure is sandwiched between the first conductive structure and the contact.

In some embodiments, a semiconductor structure is provided. The semiconductor structure includes a first source/drain structure and a second source/drain structure and nanostructures abutting the first source/drain structure and the second source/drain structure. The semiconductor structure further includes a contact landing on the first source/drain structure and a gate structure wrapping around the nanostructures. The semiconductor structure further includes a first conductive structure formed over the gate structure and a second conductive structure covering the first conductive structure and the contact. In addition, a bottom surface of a middle region of the second conductive structure is lower than a top surface of the contact.

In some embodiments, a method for manufacturing a semiconductor structure is provided. The method includes forming nanostructures over a substrate and forming a source/drain structure attached to the nanostructures. The method further includes forming a gate structure wrapping around the nanostructures and forming a contact over the source/drain structure. The method further includes forming a first conductive structure over the gate structure and

removing an upper portion of the first conductive structure to form a shortened first conductive structure. The method further includes forming a second conductive structure covering both the contact and the shortened first conductive structure.

In some embodiments, a method for manufacturing a semiconductor structure is provided. The method for manufacturing the semiconductor structure includes forming a gate structure over a substrate and forming a mask layer covering the gate structure. The method for manufacturing the semiconductor structure also includes forming a source/drain structure adjacent to the gate structure over the substrate and forming a contact over the source/drain structure. The method for manufacturing the semiconductor structure also includes forming a dielectric layer over the contact and the mask layer and forming a first trench through the dielectric layer and the mask layer over the gate structure. The method for manufacturing the semiconductor structure also includes forming a first conductive structure in the first trench and removing the upper portion of the first conductive structure. The method for manufacturing the semiconductor structure also includes forming a second conductive structure through the dielectric layer and covering the contact and the first conductive structure.

In some embodiments, a method for manufacturing a semiconductor structure is provided. The method for manufacturing the semiconductor structure includes forming nanostructures over a substrate and forming a gate structure wrapping around the nanostructures. The method for manufacturing the semiconductor structure also includes forming a source/drain structure attached to the nanostructures adjacent to the gate structure and forming a contact landing on the source/drain structure. The method for manufacturing the semiconductor structure also includes forming a dielectric layer over the contact and the gate structure and forming a first conductive structure through the dielectric layer and covering the gate structure. The method for manufacturing the semiconductor structure also includes etching an upper portion of the first conductive structure so that a top surface of the first conductive structure becomes lower than a top surface of the dielectric layer and forming a second conductive structure through the dielectric layer to cover both the contact and the first conductive structure.

In some embodiments, a semiconductor structure is provided. The semiconductor structure includes a substrate and a gate structure formed over the substrate. The semiconductor structure also includes a mask layer formed over the gate structure and a source/drain structure formed adjacent to the gate structure over the substrate. The semiconductor structure also includes a contact formed over the source/drain structure and a first conductive structure formed through the mask layer and landing on the gate structure. The semiconductor structure also includes a second conductive structure covering the first conductive structure and the contact. In addition, the second conductive structure includes an extending portion sandwiched between an upper portion of the first conductive structure and an upper portion of the contact.

In some embodiments, a semiconductor structure is provided. The semiconductor structure includes a substrate and nanostructures suspended over the substrate. The semiconductor structure also includes a gate structure wrapping around the nanostructures and a source/drain structure attached to the nanostructures. The semiconductor structure also includes a contact vertically over the source/drain structure and a first conductive structure vertically over the gate structure. The semiconductor structure also includes a

second conductive structure in contact with a top surface of the first conductive structure and a top surface of the contact and including an extending portion laterally sandwiched between the first conductive structure and the contact.

In some embodiments, a semiconductor structure is provided. The semiconductor structure includes a first source/drain structure and a second source/drain structure formed over a substrate and nanostructures suspended over the substrate. In addition, the nanostructures are spaced apart from each other in a first direction and interpose the first source/drain structure and the second source/drain structure in a second direction different from the first direction. The semiconductor structure also includes a contact landing on the first source/drain structure and a gate structure wrapping around the nanostructures and extending along a third direction different from the first direction and the second direction. The semiconductor structure also includes a first conductive structure formed over the gate structure and a second conductive structure extending into a space between the first conductive structure and the contact along the first direction. In addition, a top surface of the second conductive structure is higher than a top surface of the contact, and a bottom surface of the second conductive structure is lower than the top surface of the contact.

In some embodiments, a semiconductor structure is provided. The semiconductor structure includes a gate structure formed in a channel region of a substrate and a gate spacer covering a sidewall of the gate structure. The semiconductor structure also includes a mask layer formed over the gate structure and the gate spacer and a first conductive structure formed through the mask layer. The semiconductor structure also includes a second conductive structure formed adjacent to the first conductive structure and a contact formed over a source/drain region of the substrate. In addition, the gate spacer is laterally sandwiched between a first portion of the contact and the gate structure, the mask layer is laterally sandwiched between a second portion of the contact and the first conductive structure, the second conductive structure is laterally sandwiched between a third portion of the contact and the first conductive structure.

The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

What is claimed is:

1. A semiconductor structure, comprising:
 - nanostructures spaced apart from each other in a first direction;
 - a gate structure formed over and around the nanostructures;
 - a gate spacer covering a sidewall of the gate structure;
 - a source/drain structure attached to the nanostructures in a second direction;
 - a contact spaced apart from the gate structure by the gate spacer in the second direction;
 - a first conductive structure landing over the gate structure; and

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- a second conductive structure formed over the gate spacer, wherein a portion of the second conductive structure is sandwiched between the first conductive structure and the contact.
2. The semiconductor structure as claimed in claim 1, wherein the second conductive structure is in contact with the first conductive structure and the contact in the second direction.
3. The semiconductor structure as claimed in claim 1, wherein the second conductive structure extends over the first conductive structure and the contact in the second direction.
4. The semiconductor structure as claimed in claim 1, further comprising:
a mask layer sandwiched between the second conductive structure and the gate spacer in the first direction.
5. The semiconductor structure as claimed in claim 4, wherein the mask layer has a curved top surface.
6. The semiconductor structure as claimed in claim 1, further comprising:
a cap layer sandwiched between the gate structure and the first conductive structure in the first direction.
7. The semiconductor structure as claimed in claim 6, wherein the cap layer covers a top surface of the gate spacer.
8. The semiconductor structure as claimed in claim 6, wherein a top surface of the cap layer is substantially level with a top surface of the gate spacer.
9. The semiconductor structure as claimed in claim 1, wherein the second conductive structure has a first thickness over the contact, a second thickness over the first conductive structure, and a third thickness over the gate spacer in the first direction, and the first thickness, the second thickness, and the third thickness are different from each other.
10. A semiconductor structure, comprising:
a first source/drain structure and a second source/drain nanostructures abutting the first source/drain structure and the second source/drain structure;
a contact landing on the first source/drain structure;
a gate structure wrapping around the nanostructures;
a first conductive structure formed over the gate structure; and
a second conductive structure covering the first conductive structure and the contact,
wherein a bottom surface of a middle region of the second conductive structure is lower than a top surface of the contact.
11. The semiconductor structure as claimed in claim 10, wherein a top surface of the first conductive structure and a top surface of the contact are covered by the second conductive structure.

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12. The semiconductor structure as claimed in claim 10, further comprising:
a barrier layer laterally sandwiched between the contact and the second conductive structure, wherein a top surface of the barrier layer is lower than the top surface of the contact.
13. The semiconductor structure as claimed in claim 10, wherein the bottom surface of the middle region of the second conductive structure has a convex shape.
14. The semiconductor structure as claimed in claim 10, wherein a void is formed under the second conductive structure.
15. A method for manufacturing a semiconductor structure, comprising:
forming nanostructures over a substrate;
forming a source/drain structure attached to the nanostructures;
forming a gate structure wrapping around the nanostructures;
forming a contact over the source/drain structure;
forming a first conductive structure over the gate structure;
removing an upper portion of the first conductive structure to form a shortened first conductive structure; and
forming a second conductive structure covering both the contact and the shortened first conductive structure.
16. The method for manufacturing the semiconductor structure as claimed in claim 15, further comprising:
forming a mask layer over the gate structure;
forming a first trench in the mask layer; and
forming the first conductive structure in the first trench.
17. The method for manufacturing the semiconductor structure as claimed in claim 16, further comprising:
forming a second trench in the mask layer after the first conductive structure is formed; and
forming the second conductive structure in the second trench.
18. The method for manufacturing the semiconductor structure as claimed in claim 15, further comprising:
forming inner spacers between the nanostructures, wherein the second conductive structure vertically overlaps the inner spacers.
19. The method for manufacturing the semiconductor structure as claimed in claim 15, further comprising:
forming a cap layer over the gate structure; and
selectively depositing a conductive material over the cap layer to form the first conductive structure.
20. The method for manufacturing the semiconductor structure as claimed in claim 19, wherein the first conductive structure has a concave top surface.

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